ROUND 2.1 SCMISSION



THE VALUE CHAIN ARCHITECT:
DILIGENCE TO DELIVERY – MK LEARN & PLAY COMPANY

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TABLE OF CONTENTS

TABLE OF CONTENTS **2**
I/ Introduction **3**
II/ Executive Summary **3**
III/ Original planning & Pop-up scenario solution **4**
A. Original Planning 4
1. SCQA Analysis 4
2. Prioritization 6
3. Original Planning 8
B. Pop-up scenario solution 19
D1: Problem description 19
D2: Containment Action 20
D3: Root Causes Analysis 32
D4: Corrective Actions 34
D5. Supplier Liability and Compensation Mechanism for Defective RM4 40
IV/ Due diligence plan **42**
V/ Appendix **44**

I/ Introduction

MK is one of the high-class toy manufacturers in Vietnam, exporting to demanding international markets including the United States, Europe, and Australia. The company operates under a Make-to-Order (MTO) business model, with a strong focus on inventory optimization and achieving high customer order fulfillment. In this case, the objective is to develop an end-to-end supply chain plan covering production, material planning, and logistics, ensuring over 98% customer order fulfillment while simultaneously optimizing overall costs.The analysis is based on seven months of weekly demand across three markets and six products, together with production capacity from three fixture sets, bill of materials (BOM), current inventory, and pending orders, in order to balance demand fulfillment, operational feasibility, and cost efficiency.

During the planning process, MK also faced a pop-up scenario that tested the robustness of the proposed supply chain solution. Product A encountered a sudden quality issue caused by defective PCBA firmware, requiring all affected PCBA inventory at both the factory and the supplier to be placed on hold. Two recovery paths were considered: reflashing the firmware at the factory once the required equipment becomes available within three weeks, or sourcing newly updated PCBAs from the supplier, with inbound shipment by sea or by air depending on urgency and cost. This scenario extends the original case by requiring a revised cost assessment, a customer due diligence response, and an updated production schedule for the April 15 to May 15 period.

To ensure a structured and action-oriented response, this report is divided into two main parts. Part A focuses on the baseline planning process before the pop-up scenario, including production planning, material planning, and logistics planning under normal operating conditions. Part B focuses on resolving the pop-up scenario through the Toyota 8D problem-solving framework, connecting problem identification, root-cause analysis, corrective action design, and implementation planning across production, material planning, customer due diligence, and logistics.

II/ Executive Summary

Brief SCQ

MK Learn & Play Company is a Vietnamese premium toy manufacturer serving the US, Europe, and Australia under a Make-to-Order (MTO) model. The company must design an integrated supply chain plan that achieves >98% customer order fulfillment while balancing production capacity, material availability, and logistics cost. During the planning process, MK faced a pop-up quality issue in product A caused by defective PCBA firmware, forcing all affected inventory to be quarantined and creating major pressure on service level, recovery speed, and cost control.

Solutions & Recommendations

Adopt a data-driven integrated planning approach linking MPS, MRP, production, inventory, and logistics. In the baseline plan, focus on level-loaded production, safety stock for critical

materials, targeted bottleneck improvement for key fixtures, and a hybrid finished goods storage model using a small fixed base plus 3PL overflow during peak periods. For the pop-up scenario, apply a hybrid recovery strategy: quarantine affected stock immediately, use airfreight selectively for urgent high-priority orders, and recover the remaining volume through reflashing or sea-based replenishment. The analysis shows that 60% self-recovery provides the best balance between service continuity, shortage reduction, and cost efficiency.

Expected Results

The proposed plan is expected to protect service level above 98%, reduce shortage exposure at the most critical bottlenecks, and improve the company's ability to absorb short-term disruptions. It also limits unnecessary fixed-cost expansion by using targeted capacity upgrades, practical safety stock, and flexible warehouse design instead of broad CAPEX. Overall, the plan strengthens cost control, risk mitigation, and supply chain resilience, while ensuring that the most valuable customers and markets are protected first.

III/ Original planning & Pop-up scenario solution

A. Original Planning

1. SCQA Analysis

S – Situation

MK is facing growing demand in export markets, alongside increasing quality and fulfillment requirements, while operating under a Make-to-Order (MTO) model with a low inventory buffer, which increases its exposure to demand fluctuations.

The planning scope is broad: seven months of weekly demand across six products and three markets, fixture-based production capacity, BOM data, component inventory, pending orders, and logistics planning with a fulfillment target above 98%. From an operational perspective, MK's production system is constrained by having only three fixture sets with fixed weekly capacity, alongside a two-shift, six-day working schedule. Its bill of materials (BOM) includes both locally sourced and imported components, adding complexity in lead time management. In terms of logistics, MK relies on predefined shipment options, including 20ft FCL, 40ft FCL, and LCL.

C – Complication

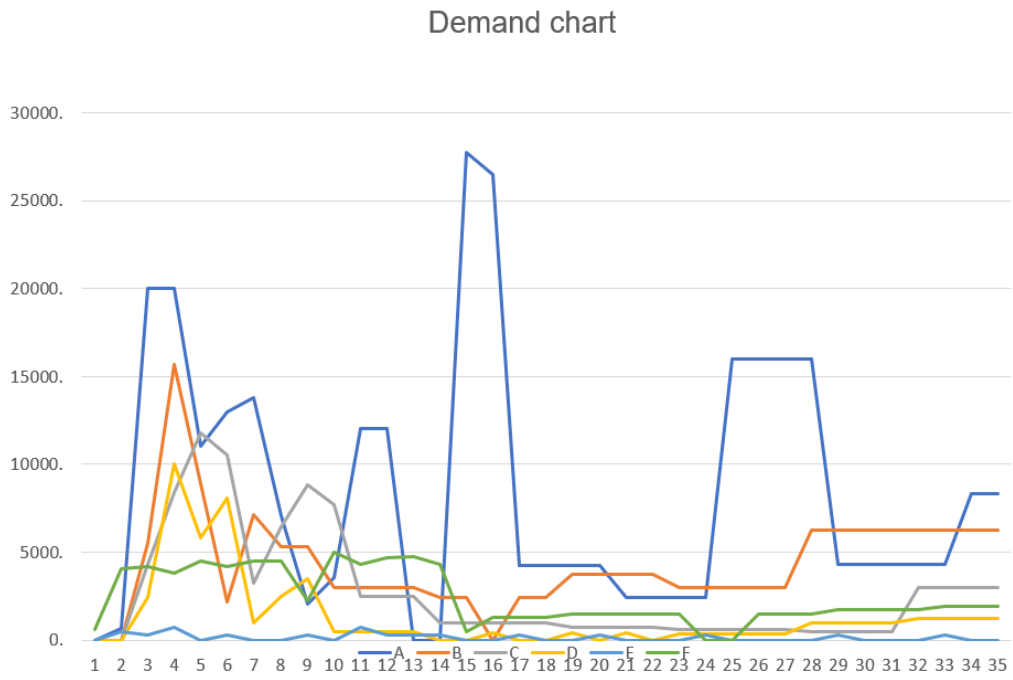


Figure 3.1: Demand chart

The supply chain already faces several structural pressures:

First, demand variability across markets is high and uneven, with sharp peaks during May–July. At the same time, multiple SKUs such as A, B, C, and D rise within a short period, creating a compound load effect on limited production and material resources. This makes capacity constraints highly binding, especially because the limited number of fixture sets reduces production flexibility.

Second , material planning is risky: BOM lead times, current inventory, and pending purchase orders reveal potential supply gaps, and without accurate reorder point (ROP) calculations, the company faces possible material shortages.

Third, there is a misalignment between production timing and shipment requirements, a misalignment between production timing and shipment requirements. Since production must be completed 1–2 weeks before shipment, fluctuating demand creates very tight and inflexible production windows.

Finally , logistics decisions involve significant trade-offs. While FCL offers lower cost per unit, it requires sufficient volume for consolidation, whereas LCL provides flexibility but at a much higher cost. Combined with carton size and container capacity constraints, there is a risk of low container utilization, which further increases logistics cost inefficiency.

Q – Question

How can MK design an integrated supply chain plan that:

- Ensures >98% customer order fulfillment
- Aligns production capacity with fluctuating demand
- Prevents material shortages via effective inventory strategy (ROP)
- Optimizes shipment planning (FCL vs LCL) to minimize logistics cost
- While managing risks in a high-uncertainty environment

A – Answer

To address these issues, MK should adopt a data-driven and fully integrated planning approach that connects production, procurement, inventory, and logistics decisions.

A Master Production Planning (MPP) approach should be implemented to smooth output across weeks through level-loading. Priority should be given to products with high demand and high variability, and during peak periods, the company should evaluate options such as overtime or selective capacity expansion to relieve bottlenecks.

In terms of material planning, MK should establish a robust inventory strategy based on reorder point (ROP) calculations that incorporate both demand variability and lead time. Safety stock should be maintained for critical components, particularly those with long lead times or high supply risk.

Production schedules should be fixed one to two weeks prior to shipment, in line with operational requirements. At the same time, selective buffer inventory can be built for high-risk SKUs to absorb demand fluctuations and reduce last-minute pressure on production.

From a logistics perspective, MK should prioritize the use of 40ft FCL shipments during high-volume periods to benefit from lower unit costs, while reserving LCL shipments only for demand gaps or urgent orders. Container utilization should be optimized based on carton CBM data to avoid inefficiencies.

In parallel, risk mitigation measures should be strengthened by identifying bottleneck fixtures early, continuously monitoring material availability, and preparing contingency plans such as alternative sourcing options or flexible shipment strategies.

In the long term, MK should focus on capability building by enhancing capacity planning processes, developing a more proactive material control system, and formalizing contingency planning for supply chain disruptions. Ultimately, the company should move toward a fully integrated, data-driven planning framework that enables coordinated decision-making across all supply chain functions, ensuring both operational efficiency and strategic resilience.

2. Prioritization

To navigate the current disruption, MK Toy utilizes ABC Analysis as a primary prioritization tool to categorize the SKU portfolio based on their cumulative financial impact. This analysis reveals that SKU A (2026 Automatic Robot) is the most critical asset, representing 57.6% of the total annual value with a high unit cost of $105. Closely following is SKU B, which

accounts for 15.8% of the total value. Together, these two SKUs constitute the "Class A" category, representing over 73% of MK Toy's revenue potential while comprising only a fraction of the total quantity. By focusing resources on these high-value items, the company ensures that recovery efforts are concentrated where the financial risk—estimated at $5.8 million for the current horizon—is greatest.

Table 3.1: ABC analysis

Item number	Annual Usage	Unit Cost	Total value	% Of Total Value	%Of Total Quantity	% Cummulative	Class
A	$288.861	105,00	30330405,0	57,6127225	45,85335422	57,6127225	A
B	$151.512	55,00	8333160,0	15,8	24,05078361	73,44159239	A
F	$70.994	90,00	6389460,0	12,1	11,26947919	81,17361703	B
C	$72.500	80,00	5800000,0	11,0	11,50853934	92,68215637	B
D	$40.930	40,00	1637200,0	3,1	6,497165725	99,17932209	C
E	$5.170	30,00	155100,0	0,3	0,820677909	100	C

To ensure maximum financial recovery during this disruption, our prioritization strategy follows a strict three-tier model based on total revenue contribution per market. Tier 1 is exclusively assigned to the US Market (SKU A), which represents a massive $22,312,500 in revenue. Because this market accounts for the vast majority of our revenue at risk, we authorize immediate Airfreight (3-day Lead Time) for new PCBA supply to guarantee 100% fulfillment and protect our most valuable market share. Tier 2 is the AU Market (SKU A), contributing $3,200,000 in revenue. Given the lower revenue density compared to the US, this tier will be managed via Internal Reflashing once the specialized equipment arrives in week three. Tier 3 is the UK Market (SKU B), with a revenue contribution of $1,430,000. For this segment, we will utilize Internal Reflashing combined with Demand Shaping—proactively offering vouchers or discounts to customers who accept a 3-week delay—ensuring that limited resources are never diverted from our higher-revenue US and AU commitments.

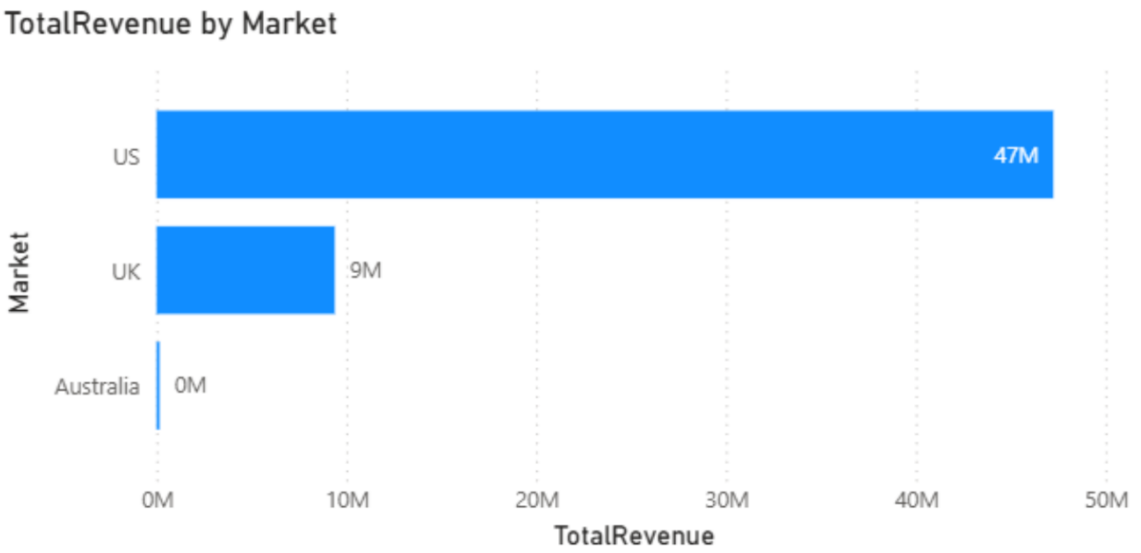


Figure 3.2: Chart of Revenue by market

The intersection of financial classification and physical production constraints dictates our Priority Order. Since the Fixture 123AB is a shared bottleneck for both Class A items (SKU A and SKU B), the production schedule must be gated by the highest value contribution. Priority 1 is strictly assigned to SKU A due to its superior unit margin and its current 62% shortage rate caused by the firmware defect. Consequently, any available Master Production Schedule (MPS) capacity on the 123AB line is first allocated to fulfill the Tier 1 US commitments for SKU A. SKU B is designated as Priority 2, acting as a production buffer during the three-week PCBA freeze to maintain factory OEE before yielding the line to SKU A's recovery phase. For secondary fixtures, such as 456CD, the same logic applies, prioritizing SKU C (Class B) over SKU D (Class C) based on their respective $80 and $40 unit costs.

3. Original Planning

3.1 MPS

3.1.1 Term explanation

Table 3.2: MPS framework explanation

Term	**Explanation**	**Formula**
Beginning FG Inventory	The physical stock of Finished Goods available at the start of the current week. It is the bridge between planning periods, representing the stock carried over from previous production.	$Beginning\ FG\ Inv_t = Ending\ FG\ Inv_{t-1}$
Carryover needed	The proactive build quantity required in the current week to cover future demand that exceeds capacity.	$= \sum max(0, Demand_{t+n} - Capacity_{t+n})$
Net required	The total quantity needed to satisfy current demand and the proactive carryover build, minus available stock.	$(Demand + Carryover\ Needed) - Opening\ FG$
Planned production	The final quantity scheduled, constrained by both the net requirement and the physical fixture limit.	$min(Net\ Required, Remaining\ shared\ fixture\ cap)$

Ending FG Inventory	Inventory remaining after fulfilling the current week's demand, which will serve the "Carryover" for the next week.	$(Opening\ FG + Planned\ production) - Demand$
Shortage	Any demand that remains unfulfilled. In this proactive model, a shortage only occurs if the entire horizon's capacity is exhausted.	$max(0, Demand - Opening\ FG - Planned\ production$
ATP (Available to promise)	Uncommitted inventory available for new orders.	$(Opening\ FG + Planned\ production) - Demand$
Capacity load (%)	The utilization rate of the fixture (e.g., 123AB).	$\frac{\sum \square\square\square\square\square\square\square\ \square\square\square\square\square\square\square\square\square\square\square(\square}{15000} * 100\%$
Shared planned before SKU	Total production already allocated to higher-priority SKUs (based on ABC analysis) on the same fixture.	$\sum Planned\ production(Priority\ 1\ SKU$
Remaining shared fixture capacity	The available production "slot" left for the current SKU after higher-priority items are scheduled.	$max(0, Weekly\ capacity - Shared\ planned\ before\ SKU)$
Fulfilled demand	The actual quantity shipped to the customer in the current week.	$min(Demand, Opening\ FG + Planned\ production)$

3.1.2 MPS Procedure to solve

- Step 1 (Define Priority & Grouping): Group SKUs by their shared fixture (e.g., A & B on 123AB) and sort them based on their ABC priority level.
- Step 2 (Horizontal Initializing): Start with Week 1. Pull the initial Beginning FG Inventory and calculate the Carryover Needed based on future demand peaks to ensure level-loading.
- Step 3 (Determine Net Needs): Calculate the Net Required for the highest-priority SKU to see if production is necessary to meet current or future smoothed demand.
- Step 4 (Sequential Capacity Allocation): For the Priority 1 SKU, the Shared Planned Before SKU is 0. For Priority 2, subtract Priority 1 production from the 15,000 total capacity to find the Remaining Shared Fixture Cap.

- Step 5 (Assign Production): Finalize Planned Production for each SKU by taking the minimum of the Net Requirement and the Remaining Capacity.
- Step 6 (Calculate Outputs): Update the Ending FG Inventory and Shortage values for the week.
- Step 7 (Iterate across Horizon): Carry the Ending FG Inventory of Week 1 to the Beginning FG Inventory of Week 2 and repeat the process for all SKUs.
- Step 8 (Bottleneck Identification): Review the Capacity Load %. Any week hitting 100% load indicates a period where demand smoothing was required to maintain a >98% service level.

3.2 MRP

A robust Material Requirements Planning (MRP) system is required to ensure component availability aligns with the finalized production plan. While the MPS manages finished goods, the MRP focuses on the "Explosion" of demand down to the raw material level.

3.2.1 Terms explanation

Table 3.3: MRP framework explanation

Term	**Explanation**	**Formula**
Gross Requirements	The total quantity of a component required to support the production start dates of all parent Finished Goods.	$\sum (Planned\ order\ release_{FG} * BOM\ usage)$
Scheduled Receipts	Pending materials currently in transit or confirmed for arrival	
Projected on hand (POH)	The estimated inventory balance at the end of each period, representing what will be available to support future production.	$POH_t = POH_{t-1} + Scheduled\ receipts_t + Planned\ order\ receipts_t - Gross\ requirement_t$
Safety Stock	A strategic buffer maintained to protect the production line against lead time variability or supply disruptions.	Local = average of demand in 1 week Overseas = average of demand in 2 weeks
Net Required	The actual quantity needed to satisfy Gross Requirements and maintain the Safety	$max(0, Gross\ req + SS - (Scheduled\ receipts + POH_{t-1}))$

	Stock buffer after consuming POH.	
Planned Order Receipts	The quantity scheduled to arrive at the factory to cover the Net Requirement gap.	
Planned Order Releases	The timing for placing orders with the supplier, calculated by offsetting the arrival date by the Logistics Lead Time.	$Planned\ order\ receipt_{t+lead\ time}$

3.2.2 Procedure to Solve

- Step 1 (Explosion of Requirements): Pull the Planned Order Releases from the MPS for all six models. Multiply these by the BOM usage to define the Gross Requirements for all sub-components.
- Step 2 (Inventory Initialization): Determine the Beginning RM Inventory and include any Scheduled Receipts from existing purchase orders.
- Step 3 (Buffer Allocation): Set the Safety Stock target based on lead time variability (e.g., 10,000 units for RM04) to ensure production continuity against supply shocks.
- Step 4 (Netting and Triggering): Calculate the Projected On-Hand for each week. If the POH is projected to fall below the Safety Stock level, a Net Requirement is triggered to trigger a new order.
- Step 5 (Logistics Offsetting): Apply the Lead-Time offset (e.g., 140 days for imported parts) to the Planned Order Receipts to determine the necessary Planned Order Releases.
- Step 6 (Final Verification): Cross-reference resulting order releases against vendor capacity and container consolidation requirements (FCL/LCL) to minimize total logistics costs.

3.3 Production planning

The production planning problem focuses on translating the baseline demand commitments into a feasible daily production schedule for the period from 15 April to 15 May, under MK's operating condition of 2 shifts per day and 6 working days per week. In line with the case brief, the production plan must ensure that finished goods are completed in time to support shipment requests while respecting the limited capacity of the three shared fixture sets. The planning logic is therefore not merely a volume allocation exercise, but a constrained prioritization problem in which multiple SKUs compete for the same production resources.

In the baseline case, before any disruption scenario occurs, the production model is built under several simplifying but explicit assumptions embedded in the solver logic. First, fixture capacity is converted from weekly capacity into daily capacity by dividing each weekly cap by six working days. Second, SKUs sharing the same fixture are allocated according to a strict

priority sequence, namely A > B, C > D, and E > F. Third, inventory is allowed to flow continuously from one shipment period to the next, so any early production that is not immediately consumed becomes carry-forward finished goods inventory. Finally, the baseline model assumes no material or BOM shortages and imposes no earliest production date restriction, meaning production may start from the first day of the planning horizon if capacity is available. These assumptions are stated directly in the planning script and are fully consistent with the original, pre-disruption production-planning purpose of the case.

Optimization logic inferred from the code

The Python script formulates the baseline production plan as an **integer linear programming model**. The decision variable is the quantity of SKU $s$produced on day t, denoted by:

$$x_{s,t} \in \mathbb{Z}_{\geq 0}$$

For each demand line $i$associated with a SKU and a shipment request date, the model also defines two state variables:

$$I_i \in \mathbb{Z}_{\geq 0} \text{ and } S_i \in \mathbb{Z}_{\geq 0}$$

where $I_i$is ending finished-goods inventory after serving that shipment period, and $S_i$is the unmet quantity, or shortage, for that demand line.

Inferred objective function

The **solver** minimizes a weighted combination of shortage and ending inventory. In mathematical form, the objective can be expressed as:

$$\min Z = \sum_{i \in \mathcal{D}} \alpha_i S_i + \beta \sum_{i \in \mathcal{D}} I_i$$

where:

- $\alpha_i = 1{,}000{,}000$ for **Priority 1** SKUs,
- $\alpha_i = 10{,}000$ for **Priority 2** SKUs,
- $\beta = 0.001$ is a very small inventory holding penalty.

This objective reveals the practical planning intent very clearly. The model is designed first and foremost to avoid shortages, especially on higher-priority SKUs. The penalty gap between priority levels is deliberately large, which means the solver will always protect a Priority 1 product if a Priority 2 product on the same fixture must be sacrificed. The tiny holding-cost coefficient then acts only as a secondary preference, encouraging a near-just-in-time production pattern and preventing the model from building unnecessary early inventory when shortage performance is unchanged. In operational terms, the solver is pursuing a lexicographic logic: service protection first, inventory minimization second.

Inferred constraints

The first major constraint is the **daily fixture capacity constraint**. Because each fixture is shared by multiple SKUs, the sum of all SKUs assigned to the same fixture on a given day cannot exceed that fixture's daily capacity:

$$\sum_{s \in \mathcal{F}(f)} x_{s,t} \leq Cap_f^{day} \forall f, \forall t$$

where $Cap_f^{day} = \frac{Cap_f^{week}}{6}$. Based on the baseline file structure, this yields approximate daily caps of 2,500 units/day for fixture 123AB, 833.3 units/day for fixture 456CD, and 500 units/day for fixture 789EF. This is exactly how the code converts the weekly factory capacity into a day-level scheduling resource.

The second core constraint is the **inventory balance constraint** for each SKU across consecutive shipment periods. For the first demand period of a SKU, beginning inventory is set to zero. For subsequent periods, beginning inventory equals the ending inventory of the previous shipment period. Production accumulated between two shipment dates is then used to satisfy the current period demand:

$$I_i - S_i = B_i + \sum_{t \in T_i} x_{s(i),t} - D_i$$

where:

- B_i is beginning inventory for demand line i,
- D_i is the demand quantity of line i,
- T_i is the set of production days assigned to that shipment period,
- $s(i)$ is the SKU corresponding to demand line i.

This balance equation is especially important because it shows that the model does not optimize each order independently. Instead, it preserves the **continuity of inventory across time**, allowing early production in one period to support later shipment requests for the same SKU. That design makes the model appropriate for the case requirement of developing a one-month production plan rather than a static period-by-period allocation.

A third set of constraints is embedded in the variable definitions themselves:

$$x_{s,t}, I_i, S_i \in \mathbb{Z}_{\geq 0}$$

This non-negativity and integrality requirement ensures that the solution is operationally realistic, since production, inventory, and shortage are all expressed in discrete units rather than fractional quantities.

Interpretation of the model design

From a managerial perspective, the inferred formulation shows that the baseline production plan is not a profit-maximization model. It is a **service-driven feasibility model**. Its role is to answer a more fundamental question: given the available fixture capacities and the predefined SKU priorities, how should MK allocate daily production so that the most critical demand is protected and visible risks are surfaced early?

This is highly appropriate for the original planning stage of the case. The brief asks teams to create a one-month production plan, ensure that products are produced ahead of shipment requests, and highlight risks where resources are insufficient. Therefore, a shortage-penalty

model with explicit fixture-sharing logic is a professionally sound planning approach because it directly translates business priorities into daily manufacturing decisions.

Results from the baseline production plan

The uploaded baseline Production_plan output shows that the planning horizon contains 21 demand lines with total demand of 143,029 units. Within the April 15 to May 15 planning window, the model schedules 103,491 units of production. Of the total shortage identified in the sheet, the vast majority belongs to shipment requests that fall beyond the covered production horizon, rather than to demand due on or before 15 May.

Most importantly, for the within-horizon demand, the baseline plan is nearly fully protected. Demand lines due by 15 May total 42,524 units, and only 2,430 units remain short, meaning that the within-horizon fulfillment level is approximately 94.3%. That shortfall is concentrated in one lower-priority SKU line rather than being spread across the portfolio.

At SKU level, the strongest result is that SKU A is completely protected in the baseline plan: all of its demand captured in the planning window is produced with zero shortage. SKU E is also fully covered. On the shared fixture pairs, the trade-off becomes more visible. On fixture 123AB, capacity is clearly biased toward A over B, which is fully consistent with the penalty structure in the model. On fixture 456CD, the model strongly protects C, while D absorbs the within-horizon shortage. On fixture 789EF, the high-priority E is protected first and the remaining capacity is allocated to F.

A second critical result is that all three fixtures are effectively operated at full load across the horizon. The daily output pattern in the sheet indicates that the plan keeps 123AB at about 2,500 units/day, 789EF at 500 units/day, and 456CD at roughly 833 units/day throughout the schedule. In other words, the model is not leaving meaningful idle capacity unused; it is already pushing the system close to its physical boundary. This means that any remaining shortage is not caused by poor sequencing, but by **structural capacity limitation**.

Key insights

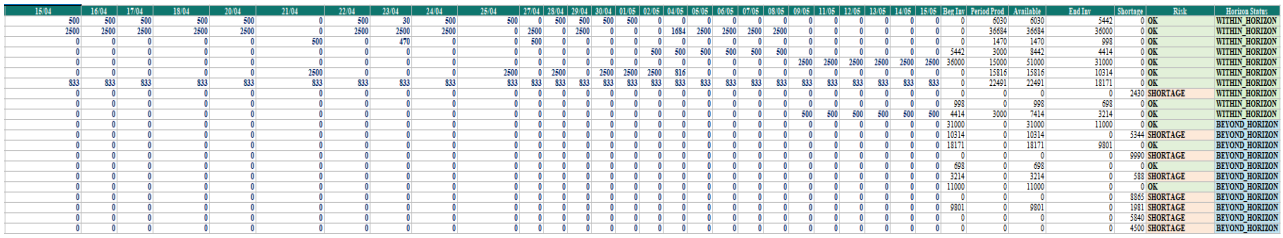
The first insight is that the baseline production problem is fundamentally a **shared-fixture prioritization problem**, not a simple one-SKU-one-line scheduling problem. Because A and B, C and D, and E and F compete for the same fixtures, the effective service outcome depends as much on the chosen priority policy as on the available capacity itself. In the current formulation, service performance for priority SKUs is achieved by deliberately allowing lower-priority products to absorb the pain.

The second insight is that the factory is already operating in a **near-saturated state** during the planning window. Since fixture loads are consistently at or near daily maximums, the remaining shortage is a structural warning sign. This indicates that any further demand increase, tighter delivery requirement, or disruption would immediately push the baseline plan into a more severe service failure.

The third insight is that the baseline plan is feasible only because it is allowed to build early inventory and carry it across shipment periods. This inventory continuity is not a side detail; it is one of the key mechanisms that makes the schedule workable. Without it, the within-horizon shortage would be materially worse.

The fourth insight is that the model provides a strong bridge between strategic and operational planning. It operationalizes the broader SCQA logic already established by the team: demand

volatility and shared fixture constraints are the core production issue, and production feasibility must be solved before logistics optimization can create value.



15/04	16/04	17/04	18/04	20/04	21/04	22/04	23/04	24/04	25/04	27/04	29/04	29/04	30/04	01/05	02/05	04/05	06/05	06/05	07/05	08/05	09/05	11/05	12/05	13/05	14/05	15/05	Elec Inv	Period Prod	Available	End Inv	Shortage	Risk	Horizon Status
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10314	0	10314	0	5344	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18171	0	18171	9801	0	OK	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9990	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	698	0	698	0	0	OK	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3214	0	3214	0	588	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11000	0	11000	0	0	OK	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8865	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9801	0	9801	0	1991	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5840	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4500	SHORTAGE	BEYOND_HORIZON

Figure X. Baseline production plan results in the Production_plan sheet

3.4 Logistics

1) Mathematical formulation

The **baseline logistics optimization problem** is formulated as a **mixed-integer linear programming (MILP)** model by **market–delivery week**, with the objective of determining the optimal transportation mix among **40ft containers, 20ft containers, and LCL shipments** so as to minimize total **ocean freight cost** while still ensuring full fulfillment of the delivery demand for each SKU. In terms of solution approach, the model is solved using an **exact MILP algorithm**, in which the solver searches for the **global optimal solution** over the set of integer decision variables through a **branch-and-bound / branch-and-cut** mechanism. As a result, the selected shipping plan is guaranteed to be the **lowest-cost solution** within the feasible region.

The model notation is defined as follows. Let $i \in I_m$denote an SKU belonging to market m, where $m \in M$is the set of markets, and $t \in T$is the delivery week. The input parameters include: d_{imt}, the demand of SKU $i$in market $m$during week t(pcs); p_i, the number of units per carton for SKU i; v_i, the volume of each carton (CBM/carton); $c_m^{40}, c_m^{20}, c_m^{LCL}$, the costs of a 40ft container, a 20ft container, and LCL shipment, respectively, for market m; and $u^{40} = 65$and $u^{20} = 28$, the CBM capacities of a 40ft container and a 20ft container, respectively. The decision variables are: x_{imt}, the number of cartons of SKU $i$shipped to market $m$in week t; $n_{mt}^{40}$and n_{mt}^{20}, the numbers of 40ft and 20ft containers used; and l_{mt}, the volume of cargo shipped via LCL, measured in CBM.

The objective function of the model is:

$$\min Z = \sum_{m \in M} \sum_{t \in T} (c_m^{40} n_{mt}^{40} + c_m^{20} n_{mt}^{20} + c_m^{LCL} l_{mt})$$

This objective function directly represents the total ocean freight cost arising from the decision on how many 40ft containers, 20ft containers, and how much LCL volume are used for each market in each week.

The model is subject to three main groups of constraints. First, the **demand fulfillment constraint** ensures that the full demand of each SKU is delivered in each week:

$$p_i x_{imt} = d_{imt} \forall i, m, t$$

Second, the **capacity constraint** ensures that the total shipment volume, after being converted into CBM, does not exceed the total capacity provided by the selected containers plus the additional LCL volume:

$$\sum_{i \in I_m} v_i \, x_{imt} \leq u^{40} n_{mt}^{40} + u^{20} n_{mt}^{20} + l_{mt} \forall m, t$$

Third, the **domain constraints** define the number of containers as non-negative integer variables, while shipment quantities and LCL volume are non-negative:

$$n_{mt}^{40}, n_{mt}^{20} \in \mathbb{Z}_+ \forall m, t$$
$$x_{imt} \geq 0, \; l_{mt} \geq 0 \forall i, m, t$$

With this structure, the model allows shipment demand to be translated directly into an optimal transportation decision, while also accurately reflecting the practical operating logic of the problem in terms of container loading by volume and ocean freight cost.

Results

Table 1 presents the **overall logistics optimization result by market**, including total shipment volume, transport mode mix, and total ocean freight cost. This table provides the main summary of the model output and shows how the optimized shipment structure differs across the US, UK, and Australia.

Market	Weeks	Total_Demand_CBM	Total_n40	Total_n20	Total_LCL_CBM	Total_Coverage_CBM	Total_Slack_CBM	Total_Cost_40	Total_Cost_20	Total_Cost_LCL	Total_Cost
AU	35	16.15625	0	0	16.15625	16.15625	0	0	0	565.43	565.43
UK	35	542.29875	2	0	412.30375	542.30375	0.005	8400	0	28861.33	37261.33
US	35	1064.867505	13	14	108.497189	1345.497189	280.629684	67600	42000	21699.42	131299.42

Markets	Weeks	Total_Demand_CBM	Total_n40	Total_n20	Total_LCL_CBM	Total_Coverage_CBM	Total_Slack_CBM	Total_Cost_40	Total_Cost_20	Total_Cost_LCL	Total_Cost
3	35	1623.322505	15	14	536.957189	1903.957189	280.634684	76000	42000	51126.18	169126.18

Table 1. Overall logistics optimization result by market

Table 2 presents the **representative weekly shipment pattern**, showing how transportation modes are selected across different weeks depending on shipment volume. Rather than showing all 35 weeks, only the most representative or peak weeks should be included in the main report so that the reader can clearly observe when the model chooses full-container shipment and when it relies on LCL.

Table 2. Representative weekly shipment pattern

Market	Week	Week_Date	Units_A	Boxes_A	CBM_A	LCL_Boxes_A	LCL_Units_A	LCL_CBM_A	Units_B	Boxes_B	CBM_B	LCL_Boxes_B	LCL_Units_B	LCL_CBM_B	d_Demand_t	n40	n20	LCL_CBM	overage_CBM	Slack_CBM	Cost_40	Cost_20	Cost_LCL	Cost	Month	Week_Date_dt
US	1	2026-04-13	0	0	0	0	0	0	10000	2500	15.625	0	0	0	15.625	0	1	0	28	12.375	0	3000	0	3000	2026-04	4/13/2026
US	2	2026-04-20	5510	1378	13.78	58	232	0.58	9490	2373	14.83125	5	20	0.03125	28.61125	0	1	0.61125	28.61125	0	0	3000	122.25	3122.25	2026-04	4/20/2026
US	3	2026-04-27	7000	1750	17.5	200	800	2	8000	2000	12.5	0	0	0	30	0	1	2	30	0	0	3000	400	3400	2026-04	4/27/2026
US	4	2026-05-04	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-05	5/4/2026
US	5	2026-05-11	12500	3125	31.25	325	1300	3.25	0	0	0	0	0	0	31.25	0	1	3.25	31.25	0	0	3000	650	3650	2026-05	5/11/2026
US	6	2026-05-18	11000	2750	27.5	0	0	0	4000	1000	6.25	920	3680	5.75	33.75	0	1	5.75	33.75	0	0	3000	1150	4150	2026-05	5/18/2026
US	7	2026-05-25	13000	3250	32.5	760	3040	7.6	2000	500	3.125	4	16	0.025	35.625	0	1	7.625	35.625	0	0	3000	1525	4525	2026-05	5/25/2026
US	8	2026-06-01	13773	3444	34.44	644	2576	6.44	1227	307	1.91875	307	1228	1.91875	36.35875	0	1	8.35875	36.35875	0	0	3000	1671.75	4671.75	2026-06	6/1/2026
US	9	2026-06-08	7058	1765	17.65	0	0	0	5309	1328	8.3	0	0	0	25.95	0	1	0	28	2.05	0	3000	0	3000	2026-06	6/8/2026
US	10	2026-06-15	2040	510	5.1	510	2040	5.1	5309	1328	8.3	1328	5312	8.3	13.4	0	0	13.4	13.4	0	0	0	0	0	2680	2026-06	6/15/2026
US	11	2026-06-22	3541	886	8.86	886	3544	8.86	3081	771	4.81875	771	3084	4.81875	13.67875	0	0	13.67875	13.67875	0	0	0	0	0	2735.75	2735.75	2026-06	6/22/2026
US	12	2026-06-29	12041	3011	30.11	671	2684	6.71	2959	740	4.625	4	16	0.025	34.735	0	1	6.735	34.735	0	0	3000	1347	4347	2026-06	6/29/2026
US	13	2026-07-06	12040	3010	30.1	210	840	2.1	2960	740	4.625	740	2960	4.625	34.725	0	1	6.725	34.725	0	0	3000	1345	4345	2026-07	7/6/2026
US	14	2026-07-13	9246	2312	23.12	0	0	0	3000	750	4.6875	0	0	0	27.8075	0	1	0	28	0.1925	0	3000	0	3000	2026-07	7/13/2026
US	15	2026-07-20	5754	1439	14.39	0	0	0	4800	1200	7.5	0	0	0	21.89	0	1	0	28	6.11	0	3000	0	3000	2026-07	7/20/2026
US	16	2026-07-27	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-07	7/27/2026
US	17	2026-08-03	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-08	8/3/2026
US	18	2026-08-10	4248	1062	10.62	1062	4248	10.62	2400	600	3.75	600	2400	3.75	14.37	0	0	14.37	14.37	0	0	0	2874	2874	2026-08	8/10/2026
US	19	2026-08-17	4248	1062	10.62	1062	4248	10.62	2400	600	3.75	600	2400	3.75	14.37	0	0	14.37	14.37	0	0	0	2874	2874	2026-08	8/17/2026
US	20	2026-08-24	4249	1063	10.63	0	0	0	3750	938	5.8625	0	0	0	16.4925	0	1	0	28	11.5075	0	3000	0	3000	2026-08	8/24/2026
US	21	2026-08-31	4248	1062	10.62	0	0	0	3750	938	5.8625	0	0	0	16.4825	0	1	0	28	11.5175	0	3000	0	3000	2026-08	8/31/2026
US	22	2026-09-07	2391	598	5.98	598	2392	5.98	3750	938	5.8625	938	3752	5.8625	11.8425	0	0	11.8425	11.8425	0	0	0	2368.5	2368.5	2026-09	9/7/2026
US	23	2026-09-14	2392	598	5.98	598	2392	5.98	3850	963	6.01875	963	3852	6.01875	11.99875	0	0	11.99875	11.99875	0	0	0	2399.75	2399.75	2026-09	9/14/2026
US	24	2026-09-21	2391	598	5.98	0	0	0	12609	3153	19.70625	0	0	0	25.68625	0	1	0	28	2.31375	0	3000	0	3000	2026-09	9/21/2026
US	25	2026-09-28	6459	1615	16.15	110	440	1.1	8291	2073	12.95625	1	4	0.00625	29.10625	0	1	1.10625	29.10625	0	0	3000	221.25	3221.25	2026-09	9/28/2026
US	26	2026-10-05	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-10	10/5/2026
US	27	2026-10-12	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-10	10/12/2026
US	28	2026-10-19	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-10	10/19/2026
US	29	2026-10-26	15000	3750	37.5	950	3800	9.5	0	0	0	0	0	0	37.5	0	1	9.5	37.5	0	0	3000	1900	4900	2026-10	10/26/2026
US	30	2026-11-02	4281	1071	10.71	0	0	0	6250	1563	9.76875	0	0	0	20.47875	0	1	0	28	7.52125	0	3000	0	3000	2026-11	11/2/2026
US	31	2026-11-09	4282	1071	10.71	0	0	0	6250	1563	9.76875	0	0	0	20.47875	0	1	0	28	7.52125	0	3000	0	3000	2026-11	11/9/2026
US	32	2026-11-16	4281	1071	10.71	0	0	0	6250	1563	9.76875	0	0	0	20.47875	0	1	0	28	7.52125	0	3000	0	3000	2026-11	11/16/2026
US	33	2026-11-23	4282	1071	10.71	0	0	0	6250	1563	9.76875	0	0	0	20.47875	0	1	0	28	7.52125	0	3000	0	3000	2026-11	11/23/2026
US	34	2026-11-30	4281	1071	10.71	0	0	0	6250	1563	9.76875	0	0	0	20.47875	0	1	0	28	7.52125	0	3000	0	3000	2026-11	11/30/2026
US	35	2026-12-07	8326	2082	20.82	257	1028	2.57	6250	1563	9.76875	3	12	0.01875	30.58875	0	1	2.58875	30.58875	0	0	3000	517.75	3517.75	2026-12	12/7/2026
US	36	2026-12-14	8325	2082	20.82	257	1028	2.57	6250	1563	9.76875	3	12	0.01875	30.58875	0	1	2.58875	30.58875	0	0	3000	517.75	3517.75	2026-12	12/14/2026

If additional evidence is needed, **Table 3** may be included in the appendix to show the **detailed load allocation by shipment**, such as which SKUs are loaded into each container or assigned to LCL. This table is not necessary for the core result discussion, but it is useful for validating that the optimization output is operationally feasible at the loading level.

Market	Week	Week_Date	Load_Type	Load_Index	Load_ID	Capacity_CBM	Boxes_A	Units_A	CBM_A	Boxes_B	Units_B	CBM_B	Boxes_F	Units_F	CBM_F	Total_Boxes	Total_Units	Used_CBM	Slack_CBM	Boxes_C	Units_C	CBM_C	Boxes_D	Units_D	CBM_D	Boxes_E	Units_E	CBM_E
US	1	2026-04-20	LCL	1	LCL		0	0.00	0.0000	0	0.00	0.0000	74	592.00	0.5781	74	592.00	0.5781	0.0000									
US	2	2026-04-27	LCL	1	LCL		171	684.00	1.7100	0	0.00	0.0000	504	4032.00	3.9375	675	4716.00	5.6475	0.0000									
US	3	2026-05-04	40	1	40_1	65.0000	5000	20000.00	50.0000	1376	5504.00	8.6000	525	4200.00	4.1016	6901	29704.00	62.7016	2.2984									
US	4	2026-05-11	40	1	40_1	65.0000	4050	16200.00	40.5000	3915	15660.00	24.4688	4	32.00	0.0313	7969	31892.00	65.0000	0.0000									
US	4	2026-05-11	LCL	1	LCL		950	3800.00	9.5000	0	0.00	0.0000	472	3776.00	3.6875	1422	7576.00	13.1875	0.0000									
US	5	2026-05-18	40	1	40_1	65.0000	2750	11000.00	27.5000	2217	8868.00	13.8563	563	4504.00	4.3984	5530	24372.00	45.7547	19.2453									
US	6	2026-05-25	40	1	40_1	65.0000	3250	13000.00	32.5000	537	2148.00	3.3563	525	4200.00	4.1016	4312	19348.00	39.9578	25.0422									
US	7	2026-06-01	40	1	40_1	65.0000	3444	13776.00	34.4400	1781	7124.00	11.1313	563	4504.00	4.3984	5788	25404.00	49.9697	15.0303									
US	8	2026-06-08	20	1	20_1	28.0000	1765	7060.00	17.6500	952	3808.00	5.9500	563	4504.00	4.3984	3280	15372.00	27.9984	0.0016									
US	8	2026-06-08	LCL	1	LCL		0	0.00	0.0000	376	1504.00	2.3500	0	0.00	0.0000	376	1504.00	2.3500	0.0000									
US	9	2026-06-15	20	1	20_1	28.0000	510	2040.00	5.1000	1328	5312.00	8.3000	275	2200.00	2.1484	2113	9552.00	15.5484	12.4516									
US	10	2026-06-22	20	1	20_1	28.0000	886	3544.00	8.8600	750	3000.00	4.6875	625	5000.00	4.8828	2261	11544.00	18.4303	9.5697									
US	11	2026-06-29	40	1	40_1	65.0000	3011	12044.00	30.1100	750	3000.00	4.6875	538	4304.00	4.2031	4299	19348.00	39.0006	25.9994									
US	12	2026-07-06	40	1	40_1	65.0000	3010	12040.00	30.1000	750	3000.00	4.6875	588	4704.00	4.5938	4348	19744.00	39.3813	25.6188									
US	13	2026-07-13	LCL	1	LCL		0	0.00	0.0000	750	3000.00	4.6875	594	4752.00	4.6406	1344	7752.00	9.3281	0.0000									
US	14	2026-07-20	LCL	1	LCL		0	0.00	0.0000	600	2400.00	3.7500	538	4304.00	4.2031	1138	6704.00	7.9531	0.0000									
US	15	2026-07-27	40	1	40_1	65.0000	6500	26000.00	65.0000	0	0.00	0.0000	0	0.00	0.0000	6500	26000.00	65.0000	0.0000									
US	15	2026-07-27	LCL	1	LCL		440	1760.00	4.4000	600	2400.00	3.7500	60	480.00	0.6888	1100	4640.00	8.6188	0.0000									
US	16	2026-08-03	40	1	40_1	65.0000	6500	26000.00	65.0000	0	0.00	0.0000	0	0.00	0.0000	6500	26000.00	65.0000	0.0000									
US	16	2026-08-03	LCL	1	LCL		122	488.00	1.2200	0	0.00	0.0000	163	1304.00	1.2734	285	1792.00	2.4934	0.0000									
US	17	2026-08-10	20	1	20_1	28.0000	1062	4248.00	10.6200	600	2400.00	3.7500	163	1304.00	1.2734	1825	7952.00	15.6434	12.3566									
US	18	2026-08-17	20	1	20_1	28.0000	1062	4248.00	10.6200	600	2400.00	3.7500	163	1304.00	1.2734	1825	7952.00	15.6434	12.3566									
US	19	2026-08-24	20	1	20_1	28.0000	1063	4252.00	10.6300	938	3752.00	5.8625	189	1512.00	1.4766	2190	9516.00	17.9691	10.0309									
US	20	2026-08-31	20	1	20_1	28.0000	1062	4248.00	10.6200	938	3752.00	5.8625	189	1512.00	1.4766	2189	9512.00	17.9591	10.0409									
US	21	2026-09-07	LCL	1	LCL		598	2392.00	5.9800	938	3752.00	5.8625	189	1512.00	1.4766	1725	7656.00	13.3191	0.0000									
US	22	2026-09-14	LCL	1	LCL		598	2392.00	5.9800	938	3752.00	5.8625	189	1512.00	1.4766	1725	7656.00	13.3191	0.0000									
US	23	2026-09-21	LCL	1	LCL		598	2392.00	5.9800	750	3000.00	4.6875	181	1448.00	1.4141	1529	6840.00	12.0816	0.0000									
US	24	2026-09-28	LCL	1	LCL		598	2392.00	5.9800	750	3000.00	4.6875	0	0.00	0.0000	1348	5392.00	10.6675	0.0000									
US	25	2026-10-05	40	1	40_1	65.0000	4005	16020.00	40.0500	750	3000.00	4.6875	0	0.00	0.0000	4755	19020.00	44.7375	20.2625									
US	26	2026-10-12	40	1	40_1	65.0000	4005	16020.00	40.0500	750	3000.00	4.6875	181	1448.00	1.4141	4936	20468.00	46.1516	18.8484									
US	27	2026-10-19	40	1	40_1	65.0000	4004	16016.00	40.0400	750	3000.00	4.6875	181	1448.00	1.4141	4935	20464.00	46.1416	18.8584									
US	28	2026-10-26	40	1	40_1	65.0000	4005	16020.00	40.0500	1563	6252.00	9.7688	181	1448.00	1.4141	5749	23720.00	51.2328	13.7672									
US	29	2026-11-02	20	1	20_1	28.0000	1071	4284.00	10.7100	1563	6252.00	9.7688	220	1760.00	1.7188	2854	12296.00	22.1975	5.8025									
US	30	2026-11-09	20	1	20_1	28.0000	1071	4284.00	10.7100	1563	6252.00	9.7688	220	1760.00	1.7188	2854	12296.00	22.1975	5.8025									
US	31	2026-11-16	20	1	20_1	28.0000	1071	4284.00	10.7100	1563	6252.00	9.7688	220	1760.00	1.7188	2854	12296.00	22.1975	5.8025									
US	32	2026-11-23	20	1	20_1	28.0000	1071	4284.00	10.7100	1563	6252.00	9.7688	220	1760.00	1.7188	2854	12296.00	22.1975	5.8025									
US	33	2026-11-30	20	1	20_1	28.0000	1071	4284.00	10.7100	1563	6252.00	9.7688	241	1928.00	1.8828	2875	12464.00	22.3616	5.6384									
US	34	2026-12-07	20	1	20_1	28.0000	2092	8328.00	20.8200	846	3384.00	5.2875	242	1936.00	1.8906	3170	13648.00	27.9981	0.0019									
US	34	2026-12-07	LCL	1	LCL		0	0.00	0.0000	717	2868.00	4.4813	0	0.00	0.0000	717	2868.00	4.4813	0.0000									
US	35	2026-12-14	20	1	20_1	28.0000	2049	8196.00	20.4900	1154	4616.00	7.2125	38	304.00	0.2969	3241	13116.00	27.9994	0.0006									
US	35	2026-12-14	LCL	1	LCL		33	132.00	0.3300	409	1636.00	2.5563	203	1624.00	1.5859	645	3392.00	4.4722	0.0000									
UK	3	2026-05-04	LCL	1	LCL											2768	6752.00	4.4000	0.0000	2160	4320.00	21.6000	608	2432.00	3.8000			
UK	4	2026-05-11	LCL	1	LCL											6683	16362.00	57.4625	0.0000	4185	8370.00	41.8500	2498	9992.00	15.6125			
UK	5	2026-05-18	40	1	40_1	65.0000										7047	17014.00	64.9950	0.0050	5587	11174.00	55.8700	1460	5840.00	9.1250			
UK	5	2026-05-18	LCL	1	LCL											304	608.00	3.0400	0.0000	304	608.00	3.0400	0	0.00	0.0000			
UK	6	2026-05-25	40	1	40_1	65.0000										7259	16566.00	65.0000	0.0000	5235	10470.00	52.3500	2024	8096.00	12.6500			
UK	6	2026-05-25	LCL	1	LCL											28	56.00	0.2763	0.0000	27	54.00	0.2700	1	4.00	0.0063			
UK	7	2026-06-01	LCL	1	LCL											1871	4244.00	17.7688	0.0000	1620	3240.00	16.2000	251	1004.00	1.5688			
UK	8	2026-06-08	LCL	1	LCL											3839	8914.00	36.0725	0.0000	3221	6442.00	32.2100	618	2472.00	3.8625			
UK	9	2026-06-15	LCL	1	LCL											5288	12282.00	49.4163	0.0000	4401	8802.00	44.0100	887	3480.00	5.4363			
UK	10	2026-06-22	LCL	1	LCL											3973	8196.00	39.2613	0.0000	3848	7696.00	38.4800	125	500.00	0.7813			
UK	11	2026-06-29	LCL	1	LCL											1375	3000.00	13.2813	0.0000	1250	2500.00	12.5000	125	500.00	0.7813			
UK	12	2026-07-06	LCL	1	LCL											1375	3000.00	13.2813	0.0000	1250	2500.00	12.5000	125	500.00	0.7813			
UK	13	2026-07-13	LCL	1	LCL											1375	3000.00	13.2813	0.0000	1250	2500.00	12.5000	125	500.00	0.7813			
UK	14	2026-07-20	LCL	1	LCL											500	1000.00	5.0000	0.0000	500	1000.00	5.0000	0	0.00	0.0000			
UK	15	2026-07-27	LCL	1	LCL											500	1000.00	5.0000	0.0000	500	1000.00	5.0000	0	0.00	0.0000			
UK	16	2026-08-03	LCL	1	LCL											600	1400.00	5.6250	0.0000	500	1000.00	5.0000	100	400.00	0.6250			
UK	17	2026-08-10	LCL	1	LCL											500	1000.00	5.0000	0.0000	500	1000.00	5.0000	0	0.00	0.0000			
UK	18	2026-08-17	LCL	1	LCL											500	1000.00	5.0000	0.0000	500	1000.00	5.0000	0	0.00	0.0000			
UK	19	2026-08-24	LCL	1	LCL											475	1150.00	4.3750	0.0000	375	750.00	3.7500	100	400.00	0.6250			
UK	20	2026-08-31	LCL	1	LCL											375	750.00	3.7500	0.0000	375	750.00	3.7500	0	0.00	0.0000			
UK	21	2026-09-07	LCL	1	LCL											475	1150.00	4.3750	0.0000	375	750.00	3.7500	100	400.00	0.6250			
UK	22	2026-09-14	LCL	1	LCL											375	750.00	3.7500	0.0000	375	750.00	3.7500	0	0.00	0.0000			
UK	23	2026-09-21	LCL	1	LCL											390	960.00	3.5625	0.0000	300	600.00	3.0000	90	360.00	0.5625			
UK	24	2026-09-28	LCL	1	LCL											390	960.00	3.5625	0.0000	300	600.00	3.0000	90	360.00	0.5625			
UK	25	2026-10-05	LCL	1	LCL											390	960.00	3.5625	0.0000	300	600.00	3.0000	90	360.00	0.5625			
UK	26	2026-10-12	LCL	1	LCL											390	960.00	3.5625	0.0000	300	600.00	3.0000	90	360.00	0.5625			
UK	27	2026-10-19	LCL	1	LCL											390	960.00	3.5625	0.0000	300	600.00	3.0000	90	360.00	0.5625			
UK	28	2026-10-26	LCL	1	LCL											500	1500.00	4.0625	0.0000	250	500.00	2.5000	250	1000.00	1.5625			
UK	29	2026-11-02	LCL	1	LCL											500	1500.00	4.0625	0.0000	250	500.00	2.5000	250	1000.00	1.5625			
UK	30	2026-11-09	LCL	1	LCL											500	1500.00	4.0625	0.0000	250	500.00	2.5000	250	1000.00	1.5625			
UK	31	2026-11-16	LCL	1	LCL											500	1500.00	4.0625	0.0000	250	500.00	2.5000	250	1000.00	1.5625			
UK	32	2026-11-23	LCL	1	LCL											1813	4252.00	16.9563	0.0000	1500	3000.00	15.0000	313	1252.00	1.9563			
UK	33	2026-11-30	LCL	1	LCL											1813	4252.00	16.9563	0.0000	1500	3000.00	15.0000	313	1252.00	1.9563			
UK	34	2026-12-07	LCL	1	LCL											1813	4252.00	16.9563	0.0000	1500	3000.00	15.0000	313	1252.00	1.9563			
UK	35	2026-12-14	LCL	1	LCL											1813	4252.00	16.9563	0.0000	1500	3000.00	15.0000	313	1252.00	1.9563			
AU	2	2026-04-27	LCL	1	LCL											238	472.00	1.4750	0.0000							238	472.00	1.4750
AU	3	2026-05-04	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	4	2026-05-11	LCL	1	LCL											349	698.00	2.1813	0.0000							349	698.00	2.1813
AU	6	2026-05-25	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	9	2026-06-15	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	11	2026-06-29	LCL	1	LCL											350	700.00	2.1875	0.0000							350	700.00	2.1875
AU	12	2026-07-06	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	13	2026-07-13	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	14	2026-07-20	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	17	2026-08-10	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	20	2026-08-31	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	24	2026-09-28	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	29	2026-11-02	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375
AU	33	2026-11-30	LCL	1	LCL											150	300.00	0.9375	0.0000							150	300.00	0.9375

Table 3: logistics all load in excel

Based on the logistics output, the optimization generates a shipment plan covering **three markets over 35 delivery weeks**, with a total shipment volume of **1,623.32 CBM** and a total ocean freight cost of **USD 169,126.18**. As shown in **Table 1**, the **US** is the dominant market in the network, accounting for **1,064.87 CBM** of shipment volume and **USD 131,299.42** in logistics cost. This means the US not only absorbs the largest share of total demand, but also drives the overall freight structure of the solution. In comparison, the **UK** accounts for **542.30 CBM** and **USD 37,261.33**, while **Australia** remains very small at only **16.16 CBM** and **USD 565.43**. Therefore, the overall result is highly concentrated, with the majority of cost and container usage driven by the US market.

Table 1 also shows that the optimized shipment mode differs substantially by market. The **US** uses a genuine **mixed shipment structure**, consisting of **13 forty-foot containers, 14 twenty-foot containers, and 108.50 CBM of LCL**, indicating that its shipment volume is both large enough to justify regular container use and variable enough to require LCL as a balancing option. The **UK**, by contrast, uses only **2 forty-foot containers** and no 20ft

containers, while the remaining **412.30 CBM** is shipped through **LCL**. This suggests that UK demand is not consistently dense enough to support frequent container loading, except in a few high-volume weeks. For **Australia**, the model selects **LCL only**, which is reasonable given the very small shipment scale and the inability to economically consolidate demand into containers.

The weekly logic of this decision can be seen more clearly in **Table 2**. This table is important because it does not merely report total cost, but explains **how** the optimization behaves from week to week. In weeks with relatively low shipment volume, the model tends to use **LCL** because it avoids paying the fixed cost of underutilized containers. In weeks with high shipment volume, especially in the US market, the model begins to select **40ft or 20ft containers**, either alone or in combination with LCL, depending on the exact weekly CBM requirement. This confirms that transportation mode selection is not fixed in advance, but is determined endogenously by the cost-minimizing logic of the MILP model.

A good example is the peak US shipment week shown in the results, where shipment volume reaches **78.19 CBM**. In that week, the model selects **one 40ft container plus 13.19 CBM of LCL**, generating a weekly cost of **USD 7,837.50**. This result is operationally meaningful: the model does not force all cargo into containers, nor does it send all cargo through LCL. Instead, it combines modes in a way that minimizes cost while fully covering required shipment volume. The same pattern can be observed in other high-volume weeks, where container use appears only when shipment scale is sufficient to justify the fixed cost of container capacity.

The result is also consistent with the freight-rate assumptions used in the model. Since the cost of a container is fixed while LCL is charged proportionally by volume, the solver compares these alternatives each week and chooses the most economical combination. This explains why some slack capacity may still appear in containerized weeks: such slack is not a modeling error, but a normal consequence of integer container decisions. In other words, the model may deliberately accept a certain amount of unused capacity if doing so is still cheaper than replacing the container with a higher amount of LCL.

B. Pop-up scenario solution

The pop-up case will be solved and analyzed by 8D Model analysis, developed by Toyota Corporation. This model has been modified in some parts to match with our scope and problem solving.

D1: Problem description

S-Situation:
A quality issue was detected in robotic A during production. Investigation confirmed a PCBA firmware defect, and all stock on hand and at vendor must be quarantined.

C-Complication:
The company now faces a supply interruption. Reflashing requires equipment that will only arrive in three weeks, while new corrected units can be produced by the supplier but must be shipped either by sea in 21 days or by air in 3 days at an added cost of USD 2 per unit. This threatens service level, production continuity, and cost performance at the same time.

Q-Question:
"How should MK redesign its short-term supply, production planning to maintain service level, minimize financial damage?"

"Howshould MK update the April 15–May 15 production schedule in a way that is operationally feasible? "

"Which recovery option is cheaper?"

"Which combination of recovery options best protects high-priority demand while containing cost and preserving the wider production system?"

A-Answer:
For the pop-up scenario, MK should respond in two steps. First, at the production-system level, defective PCBA firmware reduces A output, but because A and B share the same production line capacity, the released capacity can be shifted to B; thus, A decreases, B increases, while C, D, E, and F remain unaffected. By comparing recovery levels (%) against the combined shortage of A and B, MK finds that 60% recovery is optimal because it yields the lowest total shortage and therefore the lowest overall loss. Second, from a production planning perspective, MK should adopt a hybrid recovery strategy: 100% reflashing is too slow at about three weeks, while 100% airfreight is too costly. Therefore, MK should quarantine all affected PCBA inventory, prioritize urgent A orders for airfreight, and recover the remaining volume later through reflashing or sea-based replenishment. Under this approach, 60% self-recovery provides the best balance between service continuity, cost control, and shortage reduction.

→ Problem statement:

MK is facing an urgent quality-driven supply disruption: the robotic A model fails functional testing due to defective PCBA firmware, and all affected stock on hand and at vendor must be put on hold. Compounded by the pre-existing structural pressures, executing a recovery plan for this pop-up scenario becomes significantly more challenging. The company must decide how to recover supply and revise the April 15–May 15 production schedule while balancing cost, customer service level, and production continuity.

Decision maker(s) Supply Chain Leader with Production, Engineering, Quality, Procurement, Logistics, and Customer Service	**Criteria for successful effort** Contain defect, recover urgent supply, minimize shortage/delay/cost, maintain customer service level, deliver a feasible revised production schedule
Key forces acting on decision makers: Main trade-off: speed vs. cost. Airfreight is fast but expensive; reflashing and sea freight are cheaper but slower. Must also protect market demand and customer trust.	**Time frame for resolution** Start immediately from April 13 (date of defects discovery) with a 21-day recovery horizon
Boundaries/constraints: Affected stock unavailable; supplier capacity 4,000 units/day; 21 days by sea or 3 days by airfreight at USD 2/unit; solution must be included in the main submission.	**Accuracy necessary:** Decision-grade accuracy for cost analysis, customer due diligence, and a realistic production schedule

D2: Containment Action

In terms of pop-up scenario, MK should respond in two steps. First, at the production-system level, the defective PCBA firmware directly reduces output of product A, but because A and B share the same production line capacity, the released capacity can be shifted to B; therefore, A decreases, B increases, while C, D, E, and F remain unaffected. To determine the best operating response, MK should compare different recovery levels (%) and evaluate the combined shortage of A and B. The analysis shows that a 60% recovery level is the most suitable because it results in the lowest total shortage of A and B, and therefore the lowest overall loss.

Second, from a production planning perspective, MK should adopt a hybrid recovery strategy If MK relies on 100% reflashin , recovery takes about three weeks, which is too slow and risks losing near-term market demand. If MK relies on 100% airfreight, service can be protected, but the cost becomes too high. Therefore, MK should first quarantine all affected PCBA inventory and freeze any decisions based on uncertain stock, then split demand for product A into urgent orders, orders that can wait until the three-week recovery point, and orders that can be rescheduled with lower commercial impact. Since reflashing in three weeks and sea freight in 21 days are similar in timing, neither can rescue the most urgent demand; only airfreight in 3 days provides a real short-term recovery advantage. As a result, the best solution is to use

airfreight selectively for critical orders, while using reflashing or sea-based replenishment for the remaining volume. Under this approach, choosing 60% self-recovery gives the best balance between service continuity, cost control, and shortage reduction.

Following confirmation of the firmware defect in the PCBA used for **SKU A**, the production-planning problem changed from a normal capacity-allocation exercise into a short-term containment and recovery problem. Under the pop-up scenario, all affected PCBA stock at both the factory and the supplier had to be placed on hold, while two recovery channels became available: reflashing after the arrival of specialized equipment in approximately three weeks, or sourcing newly updated PCBAs from the supplier at a capacity of **4,000 units per day**, with inbound transport by **air in 3 days** or by **sea in 21 days**. Accordingly, the production schedule for **15 April to 15 May** could no longer be determined only by fixture capacity and SKU priority; it had to be synchronized with the time-phased availability of corrected PCBA for SKU A.

D2.1 Containment Planning Logic

The containment response was designed to achieve three objectives simultaneously. First, the plan had to protect the most urgent commitments of the disrupted SKU, especially within the immediate production horizon. Second, the plan had to keep the factory running productively despite the temporary material freeze on SKU A. Third, the plan had to minimize the premium cost of emergency inbound logistics. As a result, the updated scenario plan did not fully suspend production on the affected fixture. Instead, it temporarily redirected shared capacity toward unaffected SKUs in the early period, then gradually restored SKU A as corrected PCBA became available, and finally accelerated SKU A after the reflashed stock was released. This logic is fully consistent with the pop-up case requirement to update the production schedule, perform cost analysis, and align the response with customer service implications.

Operationally, this means the containment plan serves two purposes at once. It is a service-protection mechanism for the disrupted product and, at the same time, a throughput-preservation mechanism for the factory. The revised schedule does not attempt unlimited recovery. Rather, it selects a feasible recovery threshold embedded in the current scenario run and then determines the least-cost inbound and production pattern capable of supporting that threshold. In this sense, D2 is not merely a revised production schedule; it is a coordinated containment design linking production, inbound PCBA recovery, and emergency logistics into one decision model.

D2.2 Optimization Formulation

The scenario production model can be expressed as an integer linear program. The main decision variables are:

$$x_{s,t}$$

the production quantity of SKU $s$on workday t;

$$I_i$$

the ending finished-goods inventory for demand line i;

$$S_i$$

the shortage quantity for demand line i;

$$Air_d,\ Sea_d$$

the quantity of corrected PCBAs shipped from the supplier on calendar day $d$by air and sea, respectively.

Objective Function

In the baseline model, shortage minimization was the dominant objective. Under the pop-up scenario, the model was reformulated so that the **primary objective became the minimization of emergency airfreight cost**, while shortage penalties and inventory penalties remained in the model only as secondary terms to preserve SKU priority and discourage unnecessary early build. The objective function can therefore be written as:

$$\min\ Z = 2\sum_{d\in\mathcal{T}} Air_d + \sum_{i\in\mathcal{D}} \gamma_i S_i + \beta \sum_{i\in\mathcal{D}} I_i$$

where:

- the coefficient **2** represents the additional **USD 2 per PCBA shipped by air**;
- γ_i is a small shortage penalty used to retain the intended priority order among SKUs;
- β is a very small inventory-holding penalty, preserving a near-just-in-time preference.

This adjustment is strategically important. In the disrupted case, the key question is no longer how to eliminate shortage at any cost. The relevant question becomes: what is the minimum emergency logistics premium required to restore an acceptable service level for the disrupted product while preserving feasible output across the rest of the portfolio? The updated objective function reflects exactly that containment logic.

Constraint Set

The first group of constraints remains unchanged from the baseline model.

Fixture daily capacity constraint

$$\sum_{s\in\mathcal{F}(f)} x_{s,t} \le Cap_f^{day} \forall f, t$$

This ensures that total production on each shared fixture cannot exceed the converted daily capacity. The physical production system still operates under the same shared-fixture structure, particularly A/B on fixture 123AB, C/D on fixture 456CD, and E/F on fixture 789EF.

Inventory continuity constraint

$$I_i - S_i = B_i + \sum_{t\in T_i} x_{s(i),t} - D_i$$

This constraint preserves finished-goods continuity across shipment periods for each SKU. Ending inventory from one demand period becomes beginning inventory for the next, ensuring

that the model plans across the full operational horizon rather than treating each shipment line independently.

The second group of constraints is newly introduced by the disruption.

Supplier corrected-PCBA capacity constraint

$$Air_d + Sea_d \leq 4{,}000 \forall d$$

This reflects the supplier's maximum daily capability to produce corrected PCBAs, regardless of transport mode. Even under emergency conditions, the model cannot accelerate recovery beyond this upstream production limit.

Cumulative corrected-PCBA availability constraint for SKU A

Because each finished unit of SKU A consumes **2 PCBAs**, cumulative production of SKU A cannot exceed cumulative corrected PCBA available from airfreight, sea freight, and later reflashed stock:

$$2 \cdot \sum_{\tau=1}^{t} x_{A,\tau} \leq Reflash_t + \sum_{d:d+3\leq t} Air_d + \sum_{d:d+21\leq t} Sea_d \forall t$$

where $Reflash_t$ is zero before the reflash release date and becomes available only after the waiting period has elapsed. This is the core containment constraint because it transforms the disruption into a day-by-day synchronization rule between material recovery and factory output.

Selected service-threshold constraint for SKU A

The implemented scenario run also includes an upper bound on total allowable shortage for SKU A:

$$\sum_{i \in \mathcal{D}_A} S_i \leq 11{,}674$$

This bound should be interpreted as the selected service threshold embedded in the current scenario workbook, rather than as a requirement stated by the case brief itself. Its role is to anchor the chosen run to a minimum recovery level for SKU A, after which the model minimizes the emergency logistics premium required to satisfy that threshold.

D2.3 Revised Production Schedule

The updated production plan demonstrates a phased containment pattern rather than a simple stop-and-restart response. In the current Production_plan_new_scenario sheet, total demand remains 143,029 units, while total planned production is 103,491 units and total shortage is 39,538 units. At priority level, the scenario output reports Priority 1 shortage of 13,655 / 77,626 (17.6%) and Priority 2 shortage of 25,883 / 65,403 (39.6%), confirming that the revised plan still protects higher-priority SKUs first. In parallel, the solution uses 35,020 PCBAs by air, 0 by sea, with a total emergency airfreight premium of USD 70,040.

Inside the operational horizon, the plan remains highly targeted. The current Production_plan_new_scenario output shows that within-horizon demand totals 42,524 units, of which 40,094 units are covered, leaving only 2,430 units of shortage, equivalent to approximately 94.3% within-horizon fulfillment. Importantly, both within-horizon lines for SKU A are covered in full, while the only visible within-horizon shortage is concentrated in

SKU D. This means the containment logic successfully protects the immediate service requirements of the disrupted product and shifts the remaining loss toward lower-priority or later-period orders.

15/04	16/04	17/04	18/04	20/04	21/04	22/04	23/04	24/04	25/04	27/04	28/04	29/04	30/04	01/05	02/05	04/05	05/05	06/05	07/05	08/05	09/05	11/05	12/05	13/05	14/05	15/05	Beg Inv	Period Prod	Available	End Inv	Shortage	Risk	Horizon Status
500	500	500	500	500	500	0	500	500	500	500	500	30	500	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6030	6030	5442	0	OK	WITHIN_HORIZON
0	0	0	0	510	0	2500	0	0	2500	2500	0	0	0	2500	2000	2500	2500	2500	2500	2500	0	0	0	0	0	0	0	25010	25010	24326	0	OK	WITHIN_HORIZON
0	0	0	0	0	0	500	0	0	0	0	0	470	0	500	0	0	0	0	0	0	0	0	0	0	0	0	0	1470	1470	998	0	OK	WITHIN_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	500	500	500	500	500	500	0	0	0	0	0	0	5442	3000	8442	4414	0	OK	WITHIN_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2500	2500	2500	2500	2500	2500	24326	15000	39326	19326	0	OK	WITHIN_HORIZON
2500	2500	2500	2500	1990	2500	0	2500	2500	0	0	2500	2500	2500	0	500	0	0	0	0	0	0	0	0	0	0	0	0	27490	27490	21988	0	OK	WITHIN_HORIZON
833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	833	0	22491	22491	18171	0	OK	WITHIN_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2430	SHORTAGE	WITHIN_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	998	0	998	698	0	OK	WITHIN_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	500	500	500	500	500	500	4414	3000	7414	3214	0	OK	WITHIN_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	19326	0	19326	0	674	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	21988	0	21988	6330	0	OK	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18171	0	18171	9801	0	OK	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9990	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	698	0	698	0	0	OK	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3214	0	3214	0	588	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11000	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6330	0	6330	0	2535	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9801	0	9801	0	1981	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5840	SHORTAGE	BEYOND_HORIZON
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4500	SHORTAGE	BEYOND_HORIZON

Figure X. Revised production plan in the Production_plan_new_scenario sheet.

The figure presents the updated day-by-day allocation after the PCBA firmware disruption, including beginning inventory, period production, available quantity, ending inventory, shortage, risk status, and horizon classification for each demand line.

D2.4 Weekly Allocation Pattern

The Scenario_Weekly_Plan sheet clearly shows how the shared fixture 123AB was reallocated over time. In the week of 13 April, production of SKU A remained at 0, while SKU B absorbed 10,000 units on the same fixture. In the week of 20 April, SKU A restarted at 5,510 units, while SKU B still received 9,490 units. In the week of 27 April, SKU A increased to 7,000 units and SKU B was reduced to 8,000 units. By the week of 4 May, fixture 123AB was fully switched to 15,000 units of SKU A, and in the week of 11 May, SKU A continued at 12,500 units while SKU B stayed at zero. This pattern confirms that containment was implemented as a phased reallocation: initial protection of factory utilization through SKU B, followed by progressive restoration of SKU A once corrected PCBA became feasible.

The same weekly pattern can be observed on the other fixtures. SKU C continues to dominate fixture 456CD, while SKU D receives no production across the scenario horizon. On fixture 789EF, the plan maintains selective output for F while protecting the required volume of E. This means the containment strategy did not isolate SKU A alone; it re-optimized the entire factory under the new material gate, while still preserving the original portfolio-priority logic.

SKU	Week	Scenario Planned Production
A	13/04/2026	0
A	20/04/2026	5.510
A	27/04/2026	7.000
A	04/05/2026	15.000
A	11/05/2026	12.500
B	13/04/2026	10.000
B	20/04/2026	9.490
B	27/04/2026	8.000
B	04/05/2026	0
B	11/05/2026	0
C	13/04/2026	3.332
C	20/04/2026	4.998
C	27/04/2026	4.998
C	04/05/2026	4.998
C	11/05/2026	4.165
D	13/04/2026	0
D	20/04/2026	0
D	27/04/2026	0
D	04/05/2026	0
D	11/05/2026	0
E	13/04/2026	0
E	20/04/2026	500
E	27/04/2026	970
E	04/05/2026	0
E	11/05/2026	0
F	13/04/2026	2.000
F	20/04/2026	2.500
F	27/04/2026	2.030
F	04/05/2026	3.000
F	11/05/2026	2.500

Figure Y. Weekly production allocation in the Scenario_Weekly_Plan sheet.

The figure highlights the phased reallocation of fixture 123AB from SKU B to SKU A as corrected PCBA availability improves over the containment horizon.

D2.5 Corrected PCBA Availability Timeline

The PCBA_Supply_Timeline sheet explains why the revised production schedule takes the form above. For the first three workdays of the horizon, cumulative usable PCBA remains zero, so no production of SKU A is possible. The first corrected inbound quantity arrives by air on 18 April, allowing gradual restart thereafter. By 5 May, cumulative air-arrived corrected PCBA reaches 35,020 units. On 6 May, the reflashing option becomes available and immediately releases 370,000 corrected PCBAs, lifting cumulative usable PCBA to 405,020 units. From that point onward, the material bottleneck for SKU A is effectively removed within the horizon. The same sheet also shows that cumulative production of SKU A reaches 40,010 finished goods by 15 May, consuming 80,020 PCBAs under the 2-PCBA-per-unit BOM. Even after this consumption, the plan still retains a very large corrected-PCBA balance. This is why the model chooses no sea freight in the selected solution. Sea and reflashing share the same approximate 21-day timing, but reflashing releases a far larger volume on the same date, making sea freight economically dominated within this one-month horizon.

Workday	Date	Cal Day	Air Arrived	Sea Arrived	Reflash Avail	Cumul PCBA	A Prod Today	Cumul A Prod	PCBA Used (cum)	PCBA Remaining
1	15/04/2026	0	0	0	0	0	0	0	0	0
2	16/04/2026	1	0	0	0	0	0	0	0	0
3	17/04/2026	2	0	0	0	0	0	0	0	0
4	18/04/2026	3	4000	0	0	4000	0	0	0	4000
5	20/04/2026	5	0	0	0	4000	510	510	1020	2980
6	21/04/2026	6	0	0	0	4000	0	510	1020	2980
7	22/04/2026	7	2020	0	0	6020	2500	3010	6020	0
8	23/04/2026	8	4000	0	0	10020	0	3010	6020	4000
9	24/04/2026	9	0	0	0	10020	0	3010	6020	4000
10	25/04/2026	10	4000	0	0	14020	2500	5510	11020	3000
11	27/04/2026	12	3000	0	0	17020	2500	8010	16020	1000
12	28/04/2026	13	0	0	0	17020	0	8010	16020	1000
13	29/04/2026	14	0	0	0	17020	0	8010	16020	1000
14	30/04/2026	15	0	0	0	17020	0	8010	16020	1000
15	01/05/2026	16	4000	0	0	21020	2500	10510	21020	0
16	02/05/2026	17	4000	0	0	25020	2000	12510	25020	0
17	04/05/2026	19	4000	0	0	31020	2500	15010	30020	1000
18	05/05/2026	20	4000	0	0	35020	2500	17510	35020	0
19	06/05/2026	21	0	0	370000	405020	2500	20010	40020	365000
20	07/05/2026	22	0	0	370000	405020	2500	22510	45020	360000
21	08/05/2026	23	0	0	370000	405020	2500	25010	50020	355000
22	09/05/2026	24	0	0	370000	405020	2500	27510	55020	350000
23	11/05/2026	26	0	0	370000	405020	2500	30010	60020	345000
24	12/05/2026	27	0	0	370000	405020	2500	32510	65020	340000
25	13/05/2026	28	0	0	370000	405020	2500	35010	70020	335000
26	14/05/2026	29	0	0	370000	405020	2500	37510	75020	330000
27	15/05/2026	30	0	0	370000	405020	2500	40010	80020	325000

Figure Z. Corrected PCBA supply timeline in the PCBA_Supply_Timeline sheet.

The figure shows the early bridging role of airfreight, the release of reflashed stock on 6 May, and the synchronized ramp-up of SKU A production under cumulative material feasibility.

D2.6 Recovery and Cost Trade-off

The Recovery_Analysis sheet provides the strongest evidence for the economic logic of the chosen containment plan. As the service threshold for SKU A is tightened, airfreight increases steadily, but the overall system response is not linear. When the analysis moves from the lower recovery runs toward the current selected run, total shortage decreases sharply at first. However, beyond that point, further tightening of SKU A shortage no longer reduces total system shortage; it only reallocates scarce shared capacity from SKU B to SKU A on fixture 123AB.

This turning point is visible in the sheet. At lower recovery runs, additional airfreight materially reduces total shortage. In the current selected run, total shortage reaches 39,538 units with 35,020 PCBAs by air and USD 70,040 of airfreight cost. Beyond this point, tighter recovery runs continue to increase airfreight cost, but total shortage remains unchanged at 39,538 units. The reason is structural: once corrected PCBA is no longer the dominant constraint, the bottleneck shifts back to the shared fixture capacity of 123AB. Therefore, additional airfreight beyond the selected run does not create more total factory output; it merely changes the mix between SKU A and SKU B.

This result is essential for D2 because it demonstrates that the containment plan is not only feasible, but economically disciplined. The selected scenario run sits at the point where emergency logistics still produces real service benefit. After that point, the system stops buying shortage reduction and starts buying mainly portfolio substitution.

Recovery %	A Target	A Actual	A Short %	Total	Total	Air PCBA	Sea PCBA	Air Cost	A Planned	A Shortage	B Planned	B Shortage	C Planned	C Shortage	D Planned	D Shortage	E Planned	E Shortage	F Planned	F Shortage
0%	29184	29184	56.5%	54513	88516	0	0	0	22500	29184	30025	0	22491	1981	0	18260	1470	0	12030	5088
10%	26266	26266	50.8%	51595	91434	5836	0	11672	25418	26266	30025	0	22491	1981	0	18260	1470	0	12030	5088
20%	23348	23348	45.2%	49677	94352	11672	0	23344	28336	23348	30025	0	22491	1981	0	18260	1470	0	12030	5088
30%	20429	20429	39.5%	45758	97271	17510	0	35020	31255	20429	30025	0	22491	1981	0	18260	1470	0	12030	5088
40%	17511	17511	33.9%	42840	100189	23346	0	46692	34173	17511	30025	0	22491	1981	0	18260	1470	0	12030	5088
50%	14592	14592	28.2%	39921	103108	29184	0	58368	37092	14592	30025	0	22491	1981	0	18260	1470	0	12030	5088
60%	11674	11674	22.6%	39538	103491	35020	0	70040	40010	11674	27490	2535	22491	1981	0	18260	1470	0	12030	5088
70%	8756	8756	16.9%	39538	103491	40856	0	81712	42928	8756	24572	5453	22491	1981	0	18260	1470	0	12030	5088
80%	5837	5837	11.3%	39538	103491	46694	0	93388	45847	5837	21653	8372	22491	1981	0	18260	1470	0	12030	5088
90%	2919	2919	5.6%	39538	103491	52530	0	105060	48765	2919	18735	11290	22491	1981	0	18260	1470	0	12030	5088
95%	1460	1460	2.8%	39538	103491	55448	0	110896	50224	1460	17276	12749	22491	1981	0	18260	1470	0	12030	5088
100%	0	0	0.0%	39538	103491	58368	0	116736	51684	0	15816	14209	22491	1981	0	18260	1470	0	12030	5088

Figure AA. Recovery and cost trade-off in the Recovery_Analysis sheet.

The figure shows that additional airfreight improves total shortage only up to a certain point, after which the binding constraint shifts from corrected PCBA availability to the shared fixture capacity of 123AB.

The same conclusion is supported by the Cost_Analysis sheet. The selected scenario records 370,000 defective PCBAs on hold, reflash availability from Day 21 (6 May 2026), supplier capability of 4,000 PCBAs/day, total airfreight of 35,020 PCBAs, total sea freight of 0, and total scenario cost of USD 70,040. The production impact summary in the same sheet confirms 143,029 total demand, 103,491 total planned, 39,538 total shortage, 51,684 units of SKU A demand, and 40,010 units of SKU A planned. Taken together, these outputs show that the chosen containment plan is a controlled, quantified trade-off between service restoration and emergency cost, rather than an arbitrary operational reaction.

PCBA Scenario Cost Analysis	
Item	Value
PCBA on hold (defective)	370,000
Reflash available from	Day 21 (2026-05-06)
Supplier daily capacity	4,000 PCBA/day
Logistics Decision	
Total PCBA by AIR	35,020
Total PCBA by SEA	0
Air lead time	3 days
Sea lead time	21 days
Cost Breakdown	
Air freight cost	$70,040.00
= 35,020 PCBA x $2.0/unit	
Sea freight extra cost	$0.00
Reflash cost	$0.00 (equipment only)
TOTAL SCENARIO COST	**$70,040.00**
Production Impact	
Total demand	143,029
Total planned	103,491
Total shortage	39,538
SKU A demand	51,684
SKU A planned	40,010

Figure AB. Scenario cost summary in the Cost_Analysis sheet.

The figure summarizes defective stock on hold, reflashing availability, inbound logistics decisions, total emergency airfreight, and the resulting production impact of the selected containment plan.

D2.7 Key Findings

The first key finding is that the pop-up scenario fundamentally changed the nature of production planning. The dominant problem was no longer fixture scheduling alone; it became the synchronization of production capacity, corrected material inflow, and shipment urgency. This is why the containment model required new logistics decision variables and cumulative corrected-PCBA feasibility constraints.

The second key finding is that airfreight functions only as a bridging mechanism, not as a permanent solution. Its value lies in restarting SKU A before the reflashed stock is released. Once reflashing becomes available on 6 May, the material bottleneck is effectively relaxed for the remainder of the horizon.

The third key finding is that sea freight is not selected in the current horizon because its 21-day transit time overlaps with the reflash release date, while providing much less strategic value

than the large corrected stock unlocked through reflashing. In the present one-month containment window, sea freight therefore does not improve service timing.

The fourth key finding is that the revised schedule preserves the original portfolio-priority logic even under disruption. Immediate service protection is concentrated on SKU A and other Priority 1 items, while the residual pain is pushed toward lower-priority or beyond-horizon demand lines. The within-horizon shortage profile confirms that the containment plan is selective and strategic rather than broad and indiscriminate.

D2.8 Concluding Statement

Overall, the containment action can be evaluated as operationally feasible, economically selective, and strategically aligned with service protection. The revised model does not attempt unlimited recovery at any cost. Instead, it integrates supplier recovery capacity, inbound lead time, reflashing availability, shared fixture bottlenecks, and product priority into one coherent production-control logic. As a result, the updated Production_plan_new_scenario does not merely reflect a revised schedule; it represents a structured containment mechanism that protects urgent customer commitments, preserves near-term factory throughput, and limits emergency logistics spending to the level that still produces meaningful service recovery.

3. Revised Master Production Schedule after the PCBA Scenario

After the recovery plan at the Production Planning level had been finalized, the next step was to update the Master Production Schedule so that it accurately reflected the new supply condition of the system. This adjustment was mandatory because the pop-up scenario created a PCBA firmware defect on product A, while all related inventory on hand and at the vendor had to be put on hold. In addition, the case explicitly required the team to update the production schedule for the period from 15/04 to 15/05 under the new PCBA supply scenario.

In the post-scenario adjustment flow, PP serves as the central decision layer, while MPS becomes the weekly execution layer. In other words, the revised MPS should not be rebuilt directly from the original demand, but instead must take input from the selected PP response, including the quantity of PCBA recovered by air, sea, or reflashing, the timing when PCBA becomes available, and the recoverable output of product A in each week. This approach allows the MPS to reflect the actual production capability after the incident, rather than continuing to use the pre-scenario schedule, which is no longer feasible.

From a planning perspective, the revised MPS is recalculated by fixture-week to ensure that shared capacity remains consistently controlled. The production volume of A is first constrained by the new PCBA that can reach the factory through reflashing or replacement units from the supplier. The capacity no longer used by A in each affected week is then reallocated to the SKUs sharing the same fixture, which in turn changes the planned production, ending inventory, and shortage of the related SKU group. Therefore, the revised MPS is not only an update for product A, but a full revision of the system’s shared-capacity allocation structure.

SKU	Week	Scenario Planned Production
A	4/13/2026	0
A	4/20/2026	5,510
A	4/27/2026	7,000
A	5/4/2026	15,000
A	5/11/2026	12,500
B	4/13/2026	10,000
B	4/20/2026	9,490
B	4/27/2026	8,000
B	5/4/2026	0
B	5/11/2026	0
C	4/13/2026	3,332
C	4/20/2026	4,998
C	4/27/2026	4,998
C	5/4/2026	4,998
C	5/11/2026	4,165
D	4/13/2026	0
D	4/20/2026	0
D	4/27/2026	0
D	5/4/2026	0
D	5/11/2026	0
E	4/13/2026	0
E	4/20/2026	500
E	4/27/2026	970
E	5/4/2026	0
E	5/11/2026	0
F	4/13/2026	2,000
F	4/20/2026	2,500
F	4/27/2026	2,030
F	5/4/2026	3,000
F	5/11/2026	2,500

Table 1: Weekly Scenario Production Plan

SKU	Fixture	Week	Weekly Capacity	Total Demand	Beginning FG Inv	Storage Cap	arryover Needed	Net Required	Planned Production	Ending FG Inv	Shortage	ATP	Capacity Load %	Planner Note	ared Planned Before Sh	maining Shared Fixture Ca	Fulfilled Demand
A	123AB	4/13/2026	15,000	0	0	15,000	0	0	0	0	0	0	66.7%	Scenario PP imported	0	15000	0
A	123AB	4/20/2026	15,000	0	0	15,000	0	0	5,510	5,510	0	5,510	100.0%	Scenario PP imported	0	15000	0
A	123AB	4/27/2026	15,000	684	5,510	15,000	10,000	5,174	7,000	11,826	0	11,826	100.0%	Scenario PP imported	0	15000	684
A	123AB	5/4/2026	15,000	20,000	11,826	15,000	5,000	13,174	15,000	6,826	0	6,826	100.0%	Scenario PP imported	0	15000	20,000
A	123AB	5/11/2026	15,000	20,000	6,826	15,000	0	13,174	12,500	0	674	0	83.3%	Scenario PP imported	0	15000	19,326
A	123AB	5/18/2026	15,000	11,000	0	15,000	0	11,000	11,000	0	0	0	100.0%		0	0	15000	11,000
A	123AB	5/25/2026	15,000	13,000	0	15,000	0	13,000	13,000	0	0	0	100.0%		0	0	15000	13,000
A	123AB	6/1/2026	15,000	13,773	0	15,000	0	13,773	13,773	0	0	0	100.0%		0	0	15000	13,773
A	123AB	6/8/2026	15,000	7,058	0	15,000	0	7,058	7,058	0	0	0	82.4%		0	0	15000	7,058
A	123AB	6/15/2026	15,000	2,040	0	15,000	0	2,040	2,040	0	0	0	49.0%		0	0	15000	2,040
A	123AB	6/22/2026	15,000	3,541	0	15,000	0	3,541	3,541	0	0	0	44.1%		0	0	15000	3,541
A	123AB	6/29/2026	15,000	12,041	0	15,000	0	12,041	12,041	0	0	0	100.0%		0	0	15000	12,041
A	123AB	7/6/2026	15,000	12,040	0	15,000	0	12,040	12,040	0	0	0	100.0%		0	0	15000	12,040
A	123AB	7/13/2026	15,000	0	0	15,000	9,246	9,246	9,246	9,246	0	9,246	81.6%		0	0	15000	0
A	123AB	7/20/2026	15,000	0	9,246	15,000	24,246	15,000	5,754	15,000	0	15,000	70.4%		0	0	15000	0
A	123AB	7/27/2026	15,000	27,760	15,000	15,000	11,486	24,246	15,000	2,240	0	2,240	100.0%		0	0	15000	27,760
A	123AB	8/3/2026	15,000	26,486	2,240	15,000	0	24,246	15,000	0	9,246	0	100.0%		0	0	15000	17,240
A	123AB	8/10/2026	15,000	4,248	0	15,000	0	4,248	4,248	0	0	0	44.3%		0	0	15000	4,248
A	123AB	8/17/2026	15,000	4,248	0	15,000	0	4,248	4,248	0	0	0	44.3%		0	0	15000	4,248
A	123AB	8/24/2026	15,000	4,249	0	15,000	0	4,249	4,249	0	0	0	53.3%		0	0	15000	4,249
A	123AB	8/31/2026	15,000	4,248	0	15,000	0	4,248	4,248	0	0	0	53.3%		0	0	15000	4,248
A	123AB	9/7/2026	15,000	2,391	0	15,000	0	2,391	2,391	0	0	0	40.9%		0	0	15000	2,391
A	123AB	9/14/2026	15,000	2,392	0	15,000	0	2,392	2,392	0	0	0	41.6%		0	0	15000	2,392
A	123AB	9/21/2026	15,000	2,391	0	15,000	0	2,391	2,391	0	0	0	100.0%		0	0	15000	2,391
A	123AB	9/28/2026	15,000	2,392	0	15,000	4,067	6,459	6,459	4,067	0	4,067	98.3%		0	0	15000	2,392

Table 2: Revised MPS sample

4. Revised MRP based on the Updated MPS

Once the MPS has been updated, the MRP must also be recalculated to reflect the new material requirement profile of the system. This step is critical because MRP does not change directly from the scenario itself, but changes indirectly through the revised MPS. When the planned production quantities of each SKU by week are changed, the gross requirement of all related components and raw materials also changes accordingly. Therefore, the revised MRP is the downstream layer of the revised MPS, not an independent planning module.

In this scenario, the change in MRP is especially visible for the materials linked to A and the common materials shared among SKUs affected by capacity reallocation. If A is delayed in the early weeks but recovered later through reflashing or airfreighted PCBA, then the material demand is not eliminated, but shifted over time. This shift affects not only the gross requirement, but also the net requirement, planned order receipt, and planned order release. For materials with long lead times, such a time shift may create upstream supply risk if the purchasing plan is not updated in time.

At the same time, the revised MRP also serves to validate the feasibility of the selected PP and MPS response. A recovery plan at the finished-goods level can only be considered feasible if the downstream material plan is still capable of supporting it at the right time. Therefore, the MRP section of the report must clearly demonstrate the logic chain: PP adjusts the recovery capability of A, the revised MPS updates the weekly production schedule, and the revised MRP then recalculates the corresponding material requirement timing.

Week	A	B	C	D	E	F	RM01	RM02	RM03	RM04	RM05	RM06	RM07	RM08	RM09	RM10	RM11	RM12	RM13	RM14	RM15
	0.	10000.	3332.	0.	0.	2000.	15332.	4416.	15332.	0.	0.	25000.	15000.	9996.	1666.	0.	0.	0.	0.	10000.	4000.
4/20/2026	5510.	9490.	4998.	0.	500.	2500.	22998.	6811.5	22998.	11020.	5510.	23725.	14235.	14994.	2499.	0.	0.	1500.	2000.	12500.	5000.
4/27/2026	7000.	8000.	4998.	0.	970.	2030.	22998.	6987.75	22998.	14000.	7000.	20000.	12000.	14994.	2499.	0.	0.	2910.	3880.	10150.	4060.
5/4/2026	15000.	0.	4998.	0.	0.	3000.	22998.	6624.	22998.	30000.	15000.	0.	0.	14994.	2499.	0.	0.	0.	0.	15000.	6000.
5/11/2026	12500.	0.	4165.	0.	0.	2500.	19165.	5520.	19165.	25000.	12500.	0.	0.	12495.	2082.5	0.	0.	0.	0.	12500.	5000.
5/18/2026	11000.	4000.	5000.	0.	0.	3000.	23000.	6625.	23000.	22000.	11000.	10000.	6000.	15000.	2500.	0.	0.	0.	0.	15000.	6000.
5/25/2026	13000.	2000.	5000.	0.	300.	2700.	23000.	6737.5	23000.	26000.	13000.	5000.	3000.	15000.	2500.	0.	0.	900.	1200.	13500.	5400.
6/1/2026	13773.	1227.	5000.	0.	0.	3000.	23000.	6625.	23000.	27546.	13773.	3067.5	1840.5	15000.	2500.	0.	0.	0.	0.	15000.	6000.
6/8/2026	7058.	5309.	5000.	0.	0.	3000.	20367.	5966.75	20367.	14116.	7058.	13272.5	7963.5	15000.	2500.	0.	0.	0.	0.	15000.	6000.
6/15/2026	2040.	5309.	5000.	0.	300.	2700.	15349.	4824.75	15349.	4080.	2040.	13272.5	7963.5	15000.	2500.	0.	0.	900.	1200.	13500.	5400.
6/22/2026	3541.	3081.	5000.	0.	0.	3000.	14622.	4530.5	14622.	7082.	3541.	7702.5	4621.5	15000.	2500.	0.	0.	0.	0.	15000.	6000.
6/29/2026	12041.	2959.	2500.	500.	700.	2300.	21000.	5762.5	21000.	24082.	12041.	7397.5	4438.5	7500.	1250.	1000.	750.	2100.	2800.	11500.	4600.
7/6/2026	12040.	2960.	2500.	500.	300.	2700.	21000.	5612.5	21000.	24080.	12040.	7400.	4440.	7500.	1250.	1000.	750.	900.	1200.	13500.	5400.
7/13/2026	9246.	3000.	2500.	500.	300.	2700.	18246.	4924.	18246.	18492.	9246.	7500.	4500.	7500.	1250.	1000.	750.	900.	1200.	13500.	5400.
7/20/2026	5754.	4800.	1000.	0.	300.	2700.	14554.	3626.	14554.	11508.	5754.	12000.	7200.	3000.	500.	0.	0.	900.	1200.	13500.	5400.
7/27/2026	15000.	0.	1000.	0.	0.	476.	16476.	4309.5	16476.	30000.	15000.	0.	0.	3000.	500.	0.	0.	0.	0.	2380.	952.
8/3/2026	15000.	0.	1000.	400.	0.	1298.	17698.	4512.25	17698.	30000.	15000.	0.	0.	3000.	500.	800.	600.	0.	0.	6490.	2596.
8/10/2026	4248.	2400.	1000.	0.	300.	1300.	9248.	2474.5	9248.	8496.	4248.	6000.	3600.	3000.	500.	0.	0.	900.	1200.	6500.	2600.
8/17/2026	4248.	2400.	1000.	0.	0.	1298.	8946.	2324.25	8946.	8496.	4248.	6000.	3600.	3000.	500.	0.	0.	0.	0.	6490.	2596.
8/24/2026	4249.	3750.	750.	400.	0.	1506.	10655.	2663.	10655.	8498.	4249.	9375.	5625.	2250.	375.	800.	600.	0.	0.	7530.	3012.
8/31/2026	4248.	3750.	750.	0.	300.	1506.	10554.	2712.75	10554.	8496.	4248.	9375.	5625.	2250.	375.	0.	0.	900.	1200.	7530.	3012.
9/7/2026	2391.	3750.	750.	400.	0.	1506.	8797.	2198.5	8797.	4782.	2391.	9375.	5625.	2250.	375.	800.	600.	0.	0.	7530.	3012.
9/14/2026	2392.	3850.	750.	0.	0.	1506.	8498.	2123.75	8498.	4784.	2392.	9625.	5775.	2250.	375.	0.	0.	0.	0.	7530.	3012.
9/21/2026	2391.	12609.	600.	360.	0.	1446.	17406.	4320.75	17406.	4782.	2391.	31522.5	18913.5	1800.	300.	720.	540.	0.	0.	7230.	2892.
9/28/2026	6459.	8291.	600.	360.	300.	0.	16010.	4227.5	16010.	12918.	6459.	20727.5	12436.5	1800.	300.	720.	540.	900.	1200.	0.	0.
10/5/2026	15000.	0.	600.	360.	0.	0.	15960.	4140.	15960.	30000.	15000.	0.	0.	1800.	300.	720.	540.	0.	0.	0.	0.
10/12/2026	15000.	0.	600.	360.	0.	1448.	17408.	4321.	17408.	30000.	15000.	0.	0.	1800.	300.	720.	540.	0.	0.	7240.	2896.
10/19/2026	15000.	0.	600.	360.	0.	1448.	17408.	4321.	17408.	30000.	15000.	0.	0.	1800.	300.	720.	540.	0.	0.	7240.	2896.
10/26/2026	15000.	0.	500.	1000.	0.	1448.	17948.	4431.	17948.	30000.	15000.	0.	0.	1500.	250.	2000.	1500.	0.	0.	7240.	2896.
11/2/2026	4281.	6250.	500.	1000.	300.	1754.	14085.	3502.	14085.	8562.	4281.	15625.	9375.	1500.	250.	2000.	1500.	900.	1200.	8770.	3508.
11/9/2026	4282.	6250.	500.	1000.	0.	1756.	13788.	3352.5	13788.	8564.	4282.	15625.	9375.	1500.	250.	2000.	1500.	0.	0.	8780.	3512.
11/16/2026	4281.	6250.	500.	1000.	0.	1756.	13787.	3352.25	13787.	8562.	4281.	15625.	9375.	1500.	250.	2000.	1500.	0.	0.	8780.	3512.
11/23/2026	4282.	6250.	3000.	1250.	0.	1756.	16538.	4665.	16538.	8564.	4282.	15625.	9375.	9000.	1500.	2500.	1875.	0.	0.	8780.	3512.
11/30/2026	4281.	6250.	3000.	1250.	300.	1928.	17009.	4836.25	17009.	8562.	4281.	15625.	9375.	9000.	1500.	2500.	1875.	900.	1200.	9640.	3856.
12/7/2026	8326.	6250.	3000.	1250.	0.	1930.	20756.	5697.75	20756.	16652.	8326.	15625.	9375.	9000.	1500.	2500.	1875.	0.	0.	9650.	3860.
12/14/2026	8325.	6250.	3000.	1250.	0.	1928.	20753.	5697.25	20753.	16650.	8325.	15625.	9375.	9000.	1500.	2500.	1875.	0.	0.	9640.	3856.

Table 3: BOM demand change summary

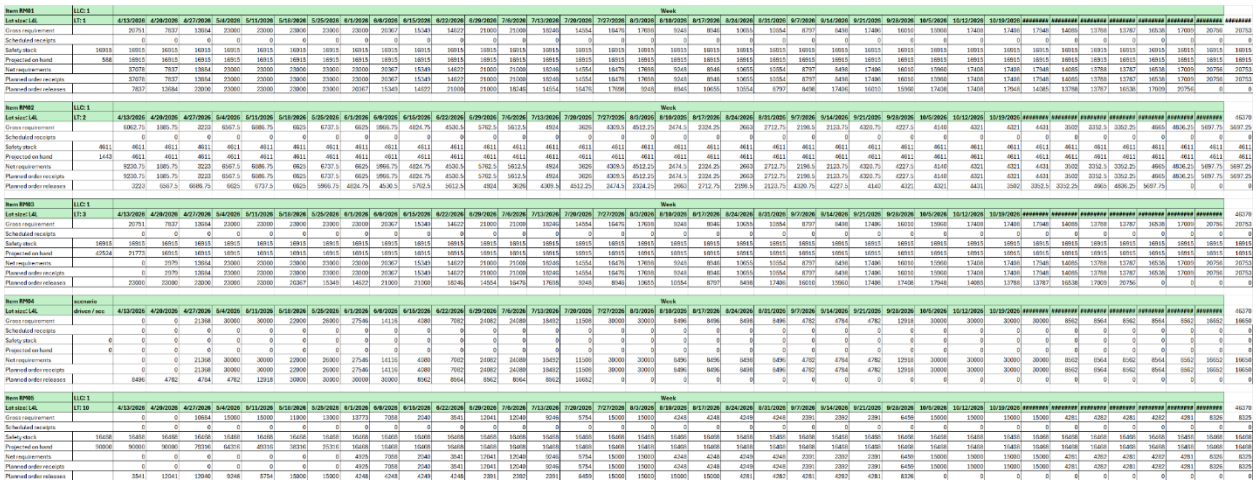


Table 4: Revised MRP sample for critical material.

5. Revised Logistics Plan after the Updated MPS

Once the revised MPS has determined the available finished-goods output by week, the logistics plan must also be rebuilt to reflect the actual shipment capability after the scenario. This is the third downstream layer after PP and MPS. If the MPS changes the output profile by week and by SKU, then the shipment profile by market, the load-building logic, and the monthly logistics cost will all change accordingly. Therefore, the post-scenario logistics plan cannot continue using the shipment plan of the base case.

In this context, the revised logistics plan must be derived from the quantity of finished goods that is truly available after incorporating shortage and after updating A's production based on the PCBA recovery timeline. These changes directly affect box volume, CBM, load type

selection, the number of containers, and the amount that must be shipped via LCL. In addition, since the scenario already introduces a special upstream cost layer through airfreighted PCBA at USD 2 per unit, the downstream logistics section should be presented separately to show that the total cost of the scenario is not limited to inbound recovery, but also extends to the outbound shipment structure.

The revised logistics plan therefore plays the role of linking the production recovery decision with the final commercial impact. When the available finished-goods output changes, not only does the service level change, but the shipment rhythm to the market also changes. This section should clearly show that post-scenario logistics is a direct consequence of the revised MPS, rather than a separate cost exercise.

Market	Week	Week_Date	Load_Type	oad_Inde	Load_ID	pacity_CB	Boxes_A	Units_A	CBM_A	Boxes_B	Units_B	CBM_B	otal_Boxe	Total_Unit	Used_CBM	Slack_CBM
US	1	2026-04-1	20	1	20_1	28	0	0	0	2500	10000	15.625	2500	10000	15.625	12.375
US	2	2026-04-2	20	1	20_1	28	1320	5280	13.2	2368	9472	14.8	3688	14752	28	0
US	2	2026-04-2	LCL	1	LCL		58	232	0.58	5	20	0.03125	63	252	0.61125	0
US	3	2026-04-2	20	1	20_1	28	1550	6200	15.5	2000	8000	12.5	3550	14200	28	0
US	3	2026-04-2	LCL	1	LCL		200	800	2	0	0	0	200	800	2	0
US	4	2026-05-0	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	4	2026-05-0	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	5	2026-05-1	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	5	2026-05-1	LCL	1	LCL		325	1300	3.25	0	0	0	325	1300	3.25	0
US	6	2026-05-1	20	1	20_1	28	2750	11000	27.5	80	320	0.5	2830	11320	28	0
US	6	2026-05-1	LCL	1	LCL		0	0	0	920	3680	5.75	920	3680	5.75	0
US	7	2026-05-2	20	1	20_1	28	2490	9960	24.9	496	1984	3.1	2986	11944	28	0
US	7	2026-05-2	LCL	1	LCL		760	3040	7.6	4	16	0.025	764	3056	7.625	0
US	8	2026-06-0	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	8	2026-06-0	LCL	1	LCL		644	2576	6.44	307	1228	1.91875	951	3804	8.35875	0
US	9	2026-06-0	20	1	20_1	28	1765	7060	17.65	1328	5312	8.3	3093	12372	25.95	2.05
US	10	2026-06-1	LCL	1	LCL		510	2040	5.1	1328	5312	8.3	1838	7352	13.4	0
US	11	2026-06-2	LCL	1	LCL		886	3544	8.86	771	3084	4.81875	1657	6628	13.67875	0
US	12	2026-06-2	20	1	20_1	28	2340	9360	23.4	736	2944	4.6	3076	12304	28	0
US	12	2026-06-2	LCL	1	LCL		671	2684	6.71	4	16	0.025	675	2700	6.735	0
US	13	2026-07-0	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	13	2026-07-0	LCL	1	LCL		210	840	2.1	740	2960	4.625	950	3800	6.725	0
US	14	2026-07-1	20	1	20_1	28	2312	9248	23.12	750	3000	4.6875	3062	12248	27.8075	0.1925
US	15	2026-07-2	20	1	20_1	28	1439	5756	14.39	1200	4800	7.5	2639	10556	21.89	6.11
US	16	2026-07-2	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	16	2026-07-2	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	17	2026-08-0	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	17	2026-08-0	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	18	2026-08-1	LCL	1	LCL		1062	4248	10.62	600	2400	3.75	1662	6648	14.37	0
US	19	2026-08-1	LCL	1	LCL		1062	4248	10.62	600	2400	3.75	1662	6648	14.37	0
US	20	2026-08-2	20	1	20_1	28	1063	4252	10.63	938	3752	5.8625	2001	8004	16.4925	11.5075
US	21	2026-08-3	20	1	20_1	28	1062	4248	10.62	938	3752	5.8625	2000	8000	16.4825	11.5175
US	22	2026-09-0	LCL	1	LCL		598	2392	5.98	938	3752	5.8625	1536	6144	11.8425	0
US	23	2026-09-1	LCL	1	LCL		598	2392	5.98	963	3852	6.01875	1561	6244	11.99875	0
US	24	2026-09-2	20	1	20_1	28	598	2392	5.98	3153	12612	19.70625	3751	15004	25.68625	2.31375
US	25	2026-09-2	20	1	20_1	28	1505	6020	15.05	2072	8288	12.95	3577	14308	28	0
US	25	2026-09-2	LCL	1	LCL		110	440	1.1	1	4	0.00625	111	444	1.10625	0
US	26	2026-10-0	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	26	2026-10-0	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	27	2026-10-1	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	27	2026-10-1	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	28	2026-10-1	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	28	2026-10-1	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	29	2026-10-2	20	1	20_1	28	2800	11200	28	0	0	0	2800	11200	28	0
US	29	2026-10-2	LCL	1	LCL		950	3800	9.5	0	0	0	950	3800	9.5	0
US	30	2026-11-0	20	1	20_1	28	1071	4284	10.71	1563	6252	9.76875	2634	10536	20.47875	7.52125
US	31	2026-11-0	20	1	20_1	28	1071	4284	10.71	1563	6252	9.76875	2634	10536	20.47875	7.52125
US	32	2026-11-1	20	1	20_1	28	1071	4284	10.71	1563	6252	9.76875	2634	10536	20.47875	7.52125
US	33	2026-11-2	20	1	20_1	28	1071	4284	10.71	1563	6252	9.76875	2634	10536	20.47875	7.52125
US	34	2026-11-3	20	1	20_1	28	1071	4284	10.71	1563	6252	9.76875	2634	10536	20.47875	7.52125
US	35	2026-12-0	20	1	20_1	28	1825	7300	18.25	1560	6240	9.75	3385	13540	28	0
US	35	2026-12-0	LCL	1	LCL		257	1028	2.57	3	12	0.01875	260	1040	2.58875	0
US	36	2026-12-1	20	1	20_1	28	1825	7300	18.25	1560	6240	9.75	3385	13540	28	0
US	36	2026-12-1	LCL	1	LCL		257	1028	2.57	3	12	0.01875	260	1040	2.58875	0

Table 6: Load building summary

Month	Market	Weeks	Total CBM	40ft Ctnr	20ft Ctnr	LCL CBM	Cost 40ft	Cost 20ft	Cost LCL	Total Cost	
2026-04	US	1	28.61	0	1	0.61	$0.00	$3,000.00	$122.25	$3,122.25	
2026-05	US	4	132.50	0	4	20.50	$0.00	$12,000.00	$4,100.00	$16,100.00	
2026-06	US	5	125.01	0	3	43.06	$0.00	$9,000.00	$8,612.50	$17,612.50	
2026-07	US	4	119.16	0	4	13.46	$0.00	$12,000.00	$2,692.00	$14,692.00	
2026-08	US	5	120.23	0	3	47.74	$0.00	$9,000.00	$9,548.00	$18,548.00	
2026-09	US	4	66.01	0	2	23.84	$0.00	$6,000.00	$4,768.25	$10,768.25	
2026-10	US	4	141.61	0	4	29.61	$0.00	$12,000.00	$5,921.25	$17,921.25	
2026-11	US	5	119.42	0	5	9.50	$0.00	$15,000.00	$1,900.00	$16,900.00	
2026-12	US	3	81.66	0	3	5.18	$0.00	$9,000.00	$1,035.50	$10,035.50	
GRAND TOTAL			**35**	**934.2**	**0**	**29**	**$193.50**	**$0.00**	**$87,000.00**	**$38,699.75**	**$125,699.75**

Table 7: Monthly logistics cost after scenario

6. Closing statement for the report

From an end-to-end planning perspective, the pop-up scenario is not only a quality issue affecting SKU A, but a shock that changes the entire short-term operating structure of the system. After the recovery plan is finalized at the PP level, the revised MPS is used to update the weekly production schedule, the revised MRP is used to shift material requirements over time, and the revised logistics plan is rebuilt to reflect the actual shipment capability after the scenario. Through this approach, the problem is resolved according to true system logic: the upstream recovery decision is consistently translated into downstream production, material, and shipment plans.

D3: Root Causes Analysis

Problem:

MK is facing a very difficult recovery process after a quality incident, because product A has been disrupted by a PCBA firmware defect. All affected inventory at both the factory and the supplier must be put on hold, while the company still needs to adjust the production schedule for the period from **15/04 to 15/05** in order to balance cost, customer service level, and production continuity.

Cause 1: Direct supply disruption for product A

Root cause 1.1: The issue lies in a core component, namely the PCBA firmware, so the affected units can no longer be used or shipped normally.

+ **Root cause 1.2:** All inventory at both the factory and the supplier has been put on hold, causing MK to lose both the current usable supply and the short-term incoming supply at the same time.
+ **Root cause 1.3:** All recovery options are structurally constrained: reflashing requires specialized equipment that will only be available after about three weeks, while replacing the PCBAs requires either sea freight with a 21-day lead time or air freight with a 3-day lead time but at higher cost.

Cause 2: The inventory structure is not strong enough to absorb disruption from both the supply side and the demand side

+ **Root cause 2.1:** About equipment , MK still has operating inventory, but there is no safety stock for equipment . As a result, when the PCBA is put on hold, the system has no protective buffer beyond normal working inventory.
+ **Root cause 2.2:** On the demand side, MK has almost no finished goods buffer, so order fulfillment depends almost entirely on current production flow. When product A is disrupted, the company has no finished goods inventory to absorb demand fluctuation or protect service level.

Cause 3: Limited backup production capacity reduces the ability to recover quickly after disruption

+ **Root cause 3.1:** The production and recovery process for product A depends on specialized equipment and processing configurations, making the system vulnerable when one critical capability is lost.
+ **Root cause 3.2:** MK has not invested sufficiently in backup machinery for critical operations, especially for technical recovery processes, so it cannot respond quickly when a disruption occurs.

Cause 4: Logistics trade-offs make the recovery both more expensive and less efficient
Root cause 4.1: Fast recovery requires air freight, but air freight significantly increases cost per unit.

+ **Root cause 4.2:** Sea freight is cheaper but too slow for demand that must be recovered in the short term, so it cannot by itself protect service level during the disruption window.
+ **Root cause 4.3:** The original logistics system already involved a trade-off between FCL and LCL in terms of load utilization, so urgent shipments make the cost structure even less efficient.

Cause 5: Planning complexity is high because the disruption hits a system that is already under pressure

+ **Root cause 5.1:** Demand across markets fluctuates significantly and unevenly, with peaks especially during the period from May to July.
+ **Root cause 5.2:** The material plan was already exposed to risk from BOM lead times and PO timing even before the pop-up occurred.

Cause 6: Customer prioritization and disruption-handling mechanisms are not yet sufficient

+ **Root cause 6.1:** When product A becomes short, MK does not yet have a ready-made system to classify customers by strategic importance, urgency of order, service risk, or business value, making it difficult to decide who should be served first under limited supply conditions.
+ **Root cause 6.3:** Customer due diligence becomes more difficult when different recovery options vary greatly in both speed and cost.

D4: Corrective Actions

Solution to root cause 2.1:

Table 3.4: Shortage of MPS baseline

SKU	Priority	Total Demand	Shortage	Short %	Revenue at Risk
D	P2	47,290	28,110	59.4%	$1,124,400
F	P2	84,558	16,646	19.7%	$1,498,140
C	P1	93,675	16,855	18.0%	$1,348,400
B	P2	151,512	17,317	11.4%	$952,435
A	P1	298,107	9,246	3.1%	$970,830
Total		**680,312**	**88,174**	**13.0%**	**$5,894,205**

The shortage pattern also supports Root Cause 2.1. The baseline MPS shows that shortages are concentrated mainly in SKUs C, D, and F rather than being evenly distributed across all products. This indicates that the problem is not a uniform plant-wide capacity shortage, but a localized vulnerability affecting specific product groups. When critical raw materials or components do not have dedicated safety stock, any disruption in those upstream inputs is transmitted directly into production shortfalls for the dependent SKUs. As a result, the system cannot absorb localized supply shocks, and shortages become concentrated in the most exposed product lines. Therefore, establishing safety stock for critical materials and components is necessary to create an upstream buffer, reduce the sensitivity of specific SKUs to disruption, and improve production continuity.

2. Bottleneck by Fixture

Fixture	Capacity/week	Shortage	Share of Total Shortage	Weeks @100%	Peak Gap	Severity
456CD	5,000	44,965	51.0%	7 weeks	+13,624/wk	Critical
789EF	3,000	16,646	18.9%	10 weeks	+2,046/wk	Medium
123AB	15,000	26,563	30.1%	6 weeks	+11,486/wk	Medium

Brief analysis:
456CD is the largest bottleneck, accounting for more than 50% of total shortage.
789EF is overloaded for a prolonged period, but the gap is not yet very large.
123AB is constrained mainly by short-term demand spikes rather than by a structural lack of base capacity.

Solution for root cause 6.1:

Solution for Root Cause 6.1:

Ưu tiên	Fixture	Vấn đề chính	Đề xuất
1	**456CD**	Thiếu công suất cấu trúc, peak demand tới 372% capacity	**Đầu tư thêm 1 fixture**, tăng cap từ 5,000 lên 10,000/tuần
2	**789EF**	Nghẽn kéo dài 10 tuần, gap khoảng 2,000/tuần	**Tăng OT 1 ca** hoặc bổ sung fixture nhỏ
3	**123AB**	Demand spike theo tuần, không thiếu công suất nền	**Không đầu tư fixture**, dùng **pre-build / carryover** trước peak

MK should establish a **priority-based capacity allocation framework** instead of investing evenly across all fixtures. **456CD** should be upgraded first because it is the main structural bottleneck, **789EF** should be addressed next through overtime or a small capacity increase, while **123AB** only requires **pre-build and carryover planning** rather than new CAPEX. This solution helps resolve Root Cause 6.1 by creating a clear rule for prioritizing limited production resources based on bottleneck severity and service impact.

Solution for root cause 2.1:

To support the production plan from the MPS without generating excessively high fixed storage costs, the company needs to determine a finished goods warehouse model that matches the actual inventory pattern. Here, the decision criterion is not only the peak inventory level, but the Beginning FG Inventory by week in the MPS, because this is the quantity that actually occupies warehouse space at the start of each period. Based on that, the company can assess whether the storage requirement is a stable base need or only short-term inventory build-up, and then choose between warehouse leasing, 3PL, or investing in its own warehouse.

1. Finished goods storage requirement based on Beginning FG Inventory from the MPS

Week	**Beginning Inv (units)**	**CBM**	**Pallets**	**Area**	**Remarks**
20/04	20,751	47.9	31.8	57.3 m²	Start of inventory build-up
27/04	28,000	59.2	39.4	70.8 m²	Entering the high range
04/05	36,500	82.7	55.0	99.0 m²	Actual warehouse peak
11/05	22,748	57.4	38.2	68.7 m²	Still high but already declining
27/07	17,400	41.2	27.4	49.4 m²	Secondary peak

05/10	19,067	33.6	22.3	40.2 m²	Final secondary peak in the horizon

Brief analysis:

The table shows that finished goods inventory does not increase evenly throughout the year, but instead concentrates in one main peak lasting about 3–4 weeks from late April to mid-May, with the peak at the week of 04/05 at around 99 m². After this period, inventory declines quite quickly and only a few additional secondary peaks appear at lower levels, around 40–50 m². This indicates that the storage requirement is temporary in nature, not a large fixed requirement throughout the entire horizon.

2. Evaluation of storage options and proposed decision

Option	**Suitability**	**Assessment**
100% short-term 3PL	Medium	Flexible but costly if used for the whole year
Long-term lease of full-size warehouse (~120 m²)	Low	Covers the peak but leaves excess capacity most of the time
Build an in-house warehouse	Very low	Actual demand is too small and not stable enough to justify CAPEX
Hybrid: small base + 3PL peak	Highest	Best fit with the inventory pattern shown in the MPS

Brief analysis:

Because the near-100 m² peak appears only for a short period, while most of the remaining weeks are significantly lower, building a warehouse or leasing a fixed full-size warehouse year-round would not be economical. The most reasonable option is to maintain a small base storage area to cover normal demand, then use 3PL overflow for the peak period from late April to early May.

3. Proposed conclusion

The company should not build a finished goods warehouse at the current stage. The most suitable option is a hybrid model: lease a fixed base area of about 40–50 m² use 3PL to cover the additional requirement during peak weeks, especially in the 20/04–11/05 cluster

Managerial conclusion:

The MPS shows that finished goods inventory increases sharply only in a few short-term weeks, so the storage decision should prioritize flexibility rather than fixed asset investment. Therefore, a small base lease + seasonal 3PL model is the most appropriate choice in both operational and cost terms.

3. Proposed safety stock for the FG warehouse based on demand data

Table 1. Safety stock sizing from weekly demand

Weekly demand metric	**Value**	**Meaning**
Average weekly demand	19,418 units	Average demand over the whole horizon
Std. dev. weekly demand	12,062 units	Large variability, but already includes seasonality
CV	62.1%	Demand is highly variable
Median weekly demand (P50)	19,736 units	“Normal” demand level
P75 weekly demand	24,128 units	High but still commonly occurring demand level
Recommended safety stock = P75 – P50	4,392 units	Practical buffer for FG

Safety stock converted into warehouse area

Safety stock conversion	Value
Safety stock	4,392 units
Volume equivalent	~13.5 CBM
Pallets equivalent	~9 pallets
Area equivalent	~16 m²

Brief analysis:

Weekly demand in the file is quite volatile, but most of this variation comes from seasonality and demand peaks that are already reflected in the Beginning FG Inventory of the MPS. Therefore, if a full statistical formula such as $z \times \sigma \times \sqrt{LT}$ is used, safety stock would become too large and would double count the seasonal inventory component. For this reason, a more appropriate level for this case is to use P75 – P50 as an "upper-normal demand buffer," resulting in about 4.4k units, equivalent to 16 m².

4. Adjustment of the warehouse decision after adding safety stock

Capacity component	Recommended area	Role
Base operating storage	40–50 m²	Covers the base FG requirement
Safety stock buffer	~16 m²	Buffer for short-term demand variation

Fixed space to lease	~55–65 m²	Reasonable long-term lease level
3PL overflow for peak	~35–45 m²	Covers the peak weeks in late April to early May
Total design capacity	~100–110 m²	

| Safe upper limit for presentation | ~120 m² | Includes peak + implementation buffer |

Brief analysis:

After adding safety stock, the core solution logic does not change: the company still should not build a warehouse and also should not lease a full-size fixed warehouse for the whole year. The difference is that the base layer should no longer stop at only 40–50 m², but should be increased to 55–65 m² in order to also hold the safety stock. At that point, the remaining peak requirement can then be shifted to 3PL overflow.

Final conclusion for the report

FG safety stock should be designed at around 4,392 units (~13.5 CBM, ~9 pallets, ~16 m²), calculated from the gap between P75 and P50 of weekly demand.
This buffer is sufficient to absorb short-term demand variation without inflating warehouse size in the way that using standard deviation over the whole horizon would. Therefore, the most reasonable option remains:
lease about 55–65 m² long-term as the base layer (including safety stock) + use 3PL for the overflow during peak weeks.

D5. Supplier Liability and Compensation Mechanism for Defective RM4

To prevent similar disruptions in the future, the company should establish a formal supplier liability and compensation mechanism for defective RM4. Since the damaged RM4 batch directly caused production loss, delivery risk, and additional operating costs, the supplier should be required to compensate for the financial losses attributable to the quality incident. This solution is not only a recovery action for the current case, but also a preventive control to reinforce supplier accountability.

Under this approach, the company should define clear contractual terms covering quality non-conformance, root-cause responsibility, compensation scope, and claim procedures. The compensation may include direct losses such as wasted material value, emergency replacement cost, overtime, rescheduling cost, delayed shipment penalties, and other proven disruption costs caused by the defective RM4. At the same time, the supplier should be placed under corrective

action monitoring, including tighter incoming inspection, mandatory CAPA submission, and stricter release conditions for future shipments.

This solution helps reduce repeated supplier quality failures in two ways. First, it creates a financial consequence for poor-quality supply, discouraging careless production or weak quality assurance from the supplier side. Second, it pushes both parties to formalize quality expectations and escalation procedures, thereby improving risk control across the upstream supply base. In short, the goal is not only to recover current losses, but also to build a stronger supplier discipline and prevention mechanism for future operations.

IV/ Due diligence plan

This plan establishes a rigorous control process to transform the uncertainties of the firmware disruption into quantifiable risks, ensuring the recovery roadmap remains on track and resilient to further shocks.

Step 1: Preparation and Planning
The preparation stage focuses on establishing strategic priorities and synchronizing internal resources. MK Toy will form a cross-functional **Crisis Management Committee** involving Production, Procurement, and Sales departments. The core objective is set: maintaining a **100% Service Level for Tier 1 (US Market)** commitments as a non-negotiable priority. During this phase, we establish a "Service Gap Budget" for Tier 3 orders and map out the resource requirements for transitioning to **24/7 triple-shift operations** on the Fixture 123AB line once the component supply is unlocked.

Step 2: Information Gathering
To navigate the uncertainty, we prioritize "Ground Truth" data over assumptions. Procurement teams will conduct an exact audit of the **340,000 quarantined RM04 units** and verify the supplier's "New Supply" output capacity (targeted at 4,000 units/day). Simultaneously, we will perform a rapid assessment of local SMT (Surface Mount Technology) vendors to evaluate the feasibility of **local reflashing outsourcing**. Real-time logistics data is gathered to compare the $2/unit Airfreight surcharge against the opportunity cost of production delays, ensuring all subsequent actions are grounded in current market reality.

Step 3: Analysis and Evaluation
In this stage, quantitative tools such as **ABC Analysis** and the **Risk Grading Matrix** are utilized to drive data-informed decisions. By identifying Model A as a **Class A item ($105/unit)**, we have quantified a total **revenue at risk of $5.8 million**. Our risk matrix indicates that "US Market Stock-out" carries the maximum risk score ($5 \times 5 = 25$ - Extremely High), which justifies the emergency Airfreight expenditure as a necessary "insurance premium." Furthermore, we evaluate the identified **62% shortage rate (74,400 units)** to determine the return on investment (ROI) for acquiring a second Fixture 123AB to accelerate the backlog burn-down.

The risk matrix is obtained as below:

Risk Event after disruption of PCBA	Likelihood (L)	Consequence (C)	Score (L*C)	Grading	Mitigation Strategy
US Market	5	5	25	Extremely	Airfreight (3-

Stock-out (Robot A)	The defect is confirmed; without intervention, production is 100% halted.	Robot A is the highest value ($105). US is the primary market. Lost revenue exceeds $100k easily.		high	day LT) for new PCBA supply to restart production immediately.
Profit Margin Erosion (Airfreight)	4 To avoid a stock-out, it is highly likely the team will authorize Airfreight.	3 $2/unit on a 10k batch is $20k. This significantly reduces the gross profit per unit but is better than a total loss.	12	Moderate	Cost-Sharing: Negotiate 50/50 split with the vendor to absorb $1/unit.
Production Idleness (Fixture 123AB)	2 Since Model B uses the same fixture, the team can pivot immediately. The risk of idleness is low.	3 If idleness occurs, fixed overheads and labor costs continue to accrue without generating output.	6	Low	Operational Pivot: Switch 100% capacity to Model B for the 3-week freeze.
Tier 3 Market Delays (UK/AU)	3 Possible because resources and components are being prioritized for Tier 1 (US).	2 Lower volume and lower price points ($30-$80) result in a minor financial impact compared to the US.	6	Low	Demand Shaping: Offer patience discounts/vouchers for delayed shipment.

Recurrence of Firmware Defect	3 Without a change in "Due Diligence" (Supplier Quality Audit), the same bug could reappear in future batches.	4 Another 20-week disruption of the best-selling SKU would be a major strategic failure.	12	Moderate	Dual souricng: to prevent reliance on a single supplier, esp for critical components Quality control: to prevent such cases happen again in the future, it requires action to standardize quality checking

Step 4: Reporting and Decision Making

The final stage establishes a **Condition-Action Framework** to ensure operational agility during the recovery window. MK Toy will implement a weekly KPI dashboard tracking production velocity, OEE (Overall Equipment Effectiveness), and Tier 1 fulfillment rates. To ensure immediate response to new uncertainties, we have established specific **"Decision Triggers"**:

- **Trigger A:** If Tier 1 Service Level falls below 98% $\rightarrow$ Immediately increase Airfreight volume.
- **Trigger B:** If backlog recovery velocity is less than 15,000 units/week $\rightarrow$ Activate mandatory weekend production shifts.
- **Trigger C:** If the risk of firmware recurrence remains high $\rightarrow$ Formally activate the **Dual-Sourcing Policy** with a local backup vendor to eliminate single-source dependency.

V/ Appendix

https://github.com/TanHongPhong/SCmission